

ChE 417 - Data, Statistics, and Modeling for Engineers and Scientists

Chapter 1 – Introduction to Data and Statistics

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Why is this course put together?

- What do you mean by statistics? - Data science or data analysis?
- Disconnect between current DS approaches and ChemE – my grad experience!
- Data-models-Decision approach

Course overview

1. Statistics – introduction and data-models-decision approach overview
2. Single variable random variables – uncertainty
3. Multivariate random variables
4. Parameter estimation for random variables – MC, probability.
5. Structural models – optimization-based approaches for parameter estimation - MLE, bayesian estimation, understanding signal from noise.
6. Advanced tools of DS and statistics – time series, regression, PCA, Lasso, NN.
7. How to make decisions in face of uncertainty – deterministic and stochastic optimization, bayesian optimization



The "Product Quality" Vector

When a factory produces a batch of material, the final product is defined by multiple physical properties that vary randomly from batch to batch due to slight changes in raw materials and machine wear.

An engineer might measure a single batch using a multivariate random vector $\mathbf{X} = [X_1, X_2]$, where:

- X_1 = Moisture content (%)
- X_2 = Particle size (μm)

Agenda

- Course overview
- Approach – slides/hands-on examples/HW
- Course syllabus
- Ethics and behavior with AI
- Github page - <https://github.com/bhalode-lab/ChE-417>

Python overview

- **Installing VS Code**
- **Reviewing Python basics –**
 - 1. Python basics (Python Overview1.ipynb, python for data modeling1 (types of libraries)
 - 2. Numpy (numpy tutorial.ipynb)
 - Renew ipynb file. Exercises 4, 5 in class. Skip 6.3 onwards.
 - 3. Pandas (pandas_introduction.ipynb) -
 - 4. Matplotlib (matplotlib_introduction.ipynb) – simple, review on your own.
 - 5. Scipy (scipy_introduction.ipynb) – to be done later.

Why do we need to understand data and statistics?

- As Engineers and Scientists, we are trained to use fundamental laws to make decisions.

- Discovery of fundamental laws – how are they devised? – result of extensive collection and analysis of observations (data).
- **Fundamental laws** - often expressed in form of mechanistic or first-principle based model.
- **Mechanistic model** – provides a concise summary of observations (knowledge) that allows us to predict and generalize.

- Decisions such as –

1. Material design
2. Design and operation of renewable energy – with production.
3. Biological processes – optimization
4. Decision to conduct more experiments or not?

Fundamental laws – dependence?

Fundamental laws are powerful but only provide limited descriptions of phenomena:

- Laws are applicable under specific settings (e.g., continuum vs. atomistic)
- Discovering laws and new mechanistic models might be challenging or cost-prohibitive (e.g., climate)

- How can we make decisions and predictions when fundamental laws are missing or without their use?

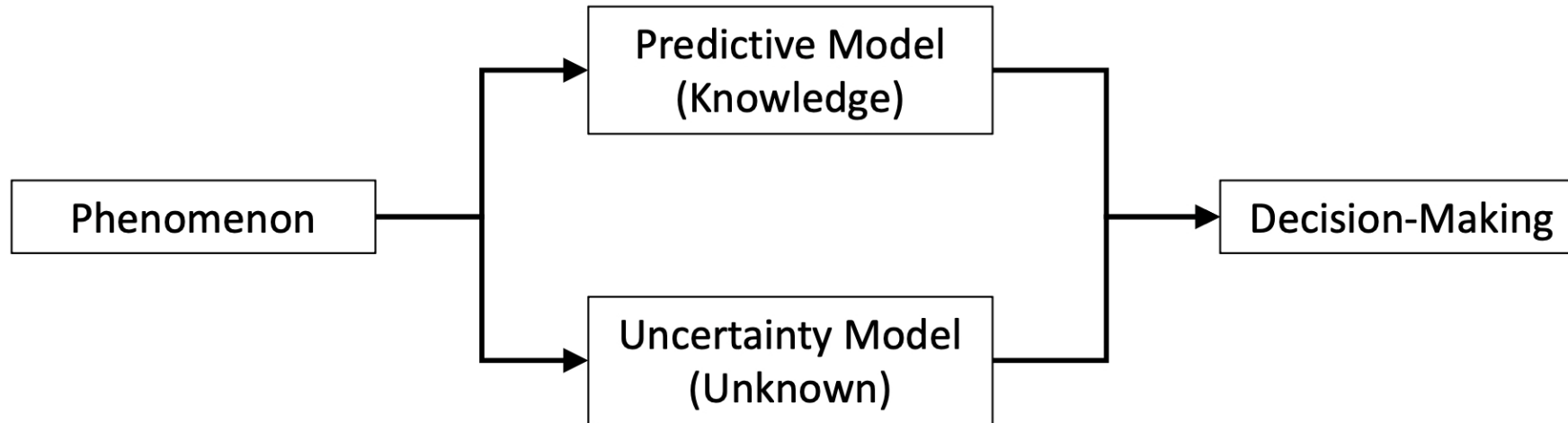
- To account for this, we often build predictive models – purely based on data – called empirical or data-driven models.
- Generally, a combination of empirical or data-driven and mechanistic knowledge is used.

Can you think of some fundamental and empirical + fundamental laws?

Mechanistic + empirical models -

- Model predictions will *always* face a certain degree of *uncertainty* (due to limited knowledge/understanding/data or due to *random* phenomena affecting systems).
 - Despite these limitations, we still need to be able to *make decisions*. In fact, we (as humans), make predictions and decisions in our daily lives.
 - The human brain naturally gathers knowledge (learns) from observations by building empirical models and has the ability to blend such empirical knowledge with mechanistic knowledge.
 - The human brain has a natural ability to hedge against uncertainty and adapts decisions based on new knowledge.
- Examples – riding a bike, cooking, baking, driving a car – navigating busy streets.

How to make decisions?



- A combination of knowledge (mechanistic and empirical – what is predictable) and characterization of unknown (uncertainty) – what is not predictable.

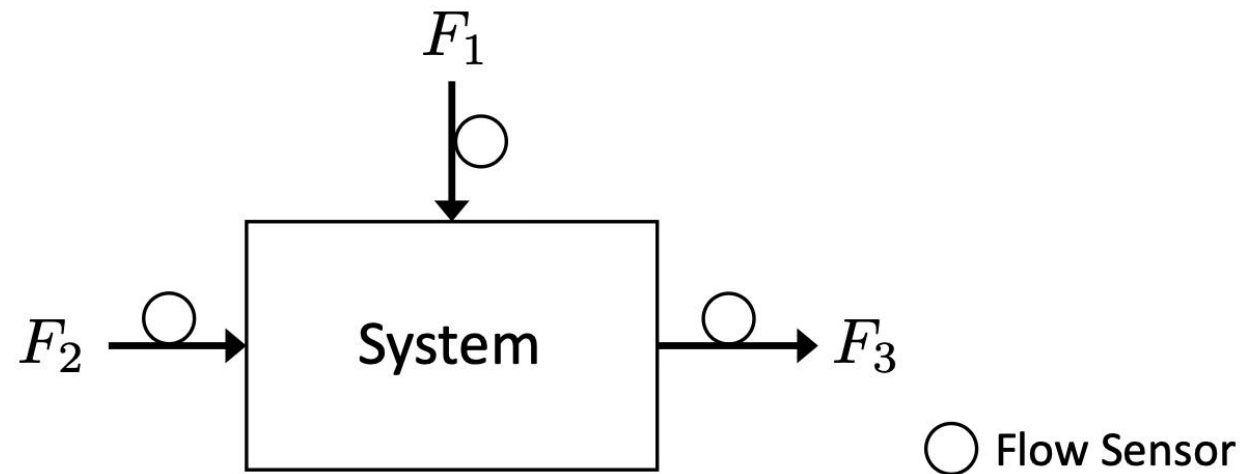
Introduction to Statistics

- Statistics is the branch of mathematics that offers tool to –
 - 1. collect, analyze and extract knowledge (models) from data
 - 2. Characterize and model the unknown (uncertainty)
 - 3. Systematically make decisions in the face of uncertainty.

For an engineering perspective, *statistics* aids the discovery of fundamental laws and the development of mechanistic and empirical models as well as the characterization of random phenomena.

Let's look at an example – flow mixer -

- Input flows – F_1 and F_2 .
- Output flows – F_3
- Sensor measurements at each location –
- $f_1(t) = 101.5$, $f_2(t) = 50.5$ and $f_3(t) = 151$ gallons per min (gpm)



Question –

Conservation law tells us that $f_3 = f_1 + f_2$ at all times. But this is not true.

Why?

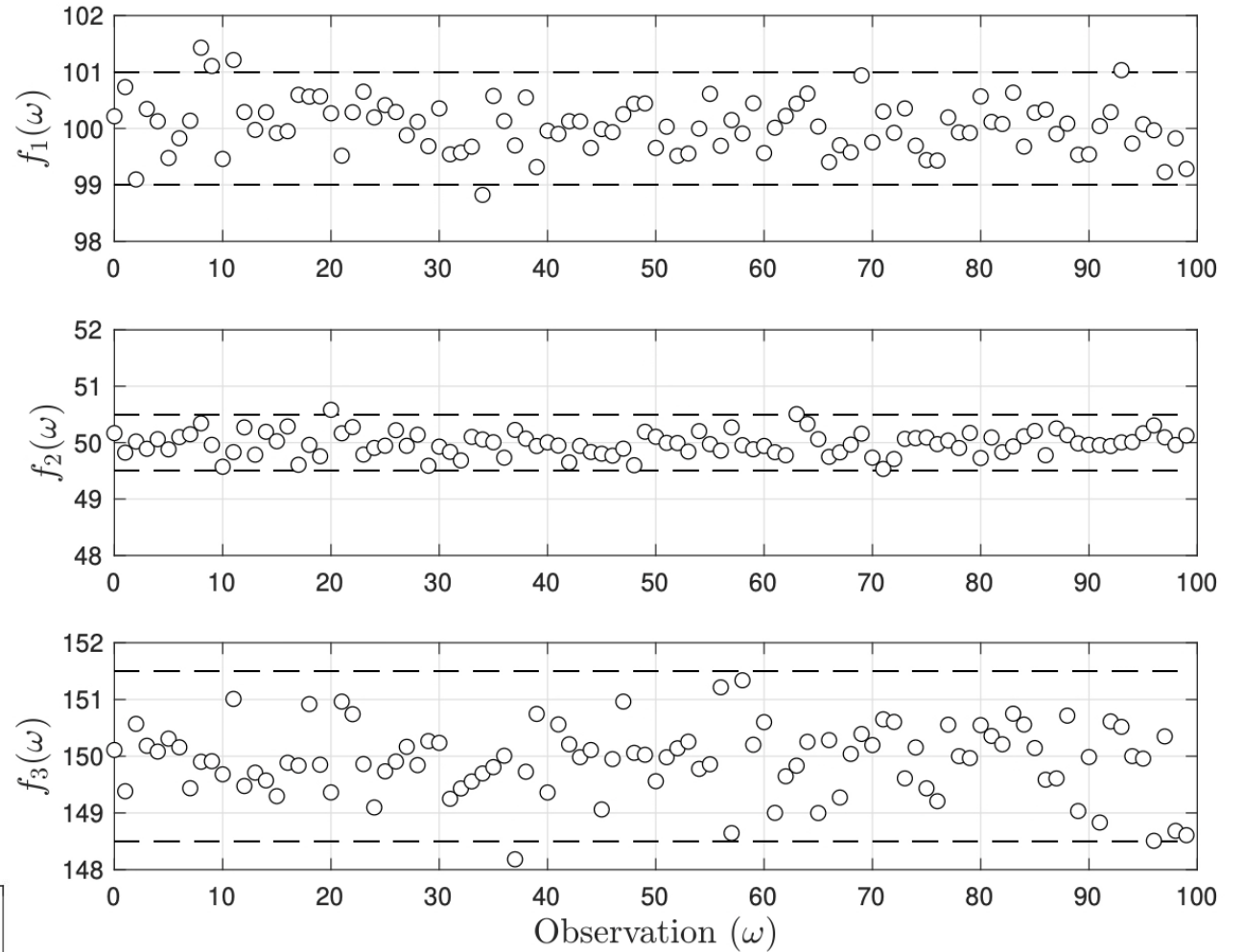
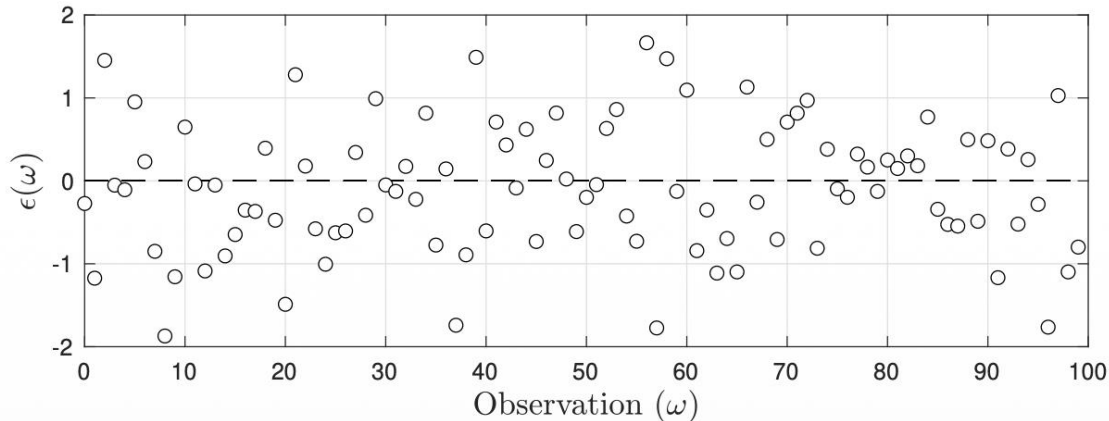
What is known and unknown about this system?

What is the source of uncertainty?

Collect data -

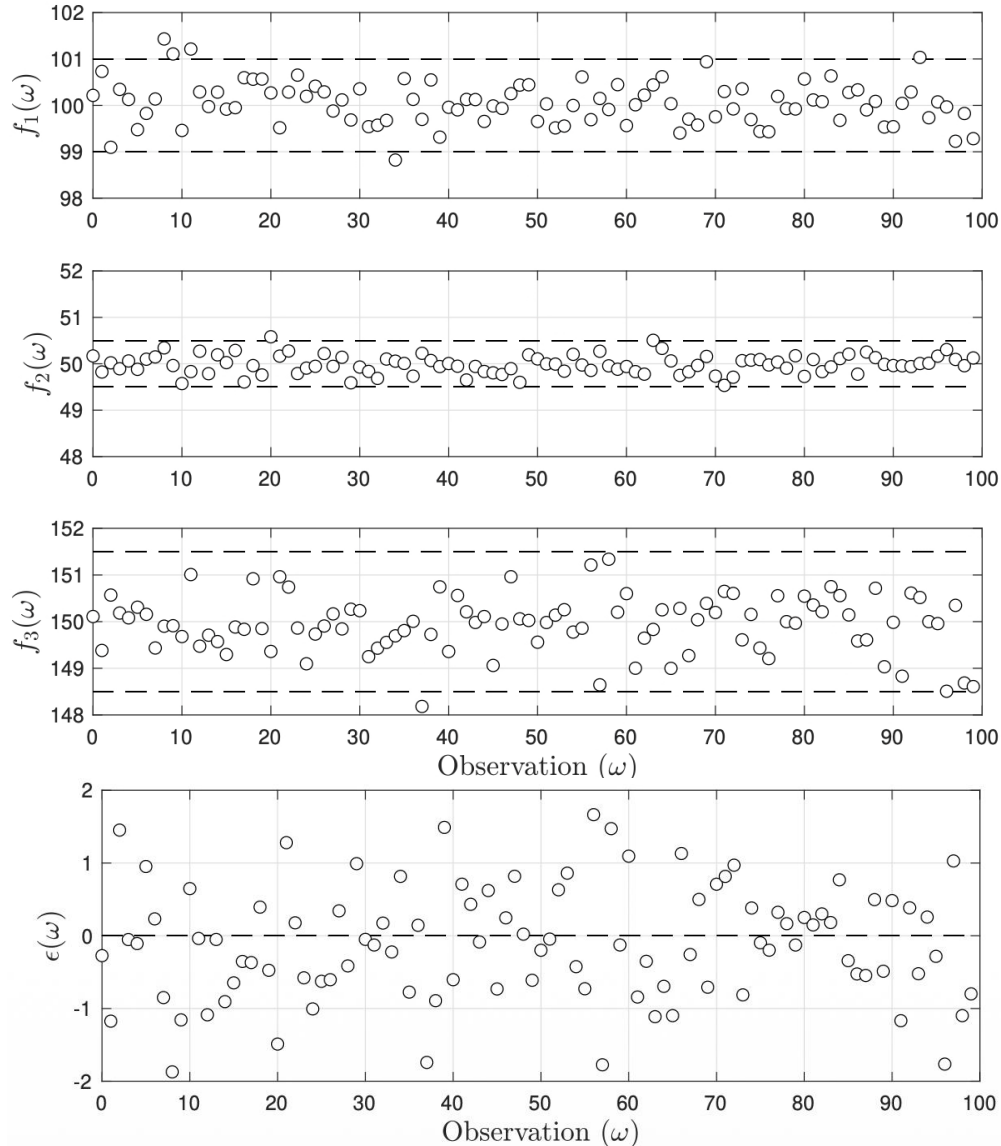
- Take 100 measurements.
- Calculate the error =

$$\epsilon(\omega) = f_3(\omega) - f_1(\omega) - f_2(\omega)$$



What is known and unknown about this system?
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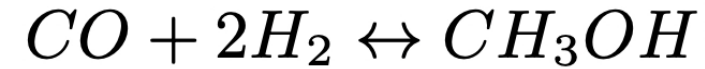
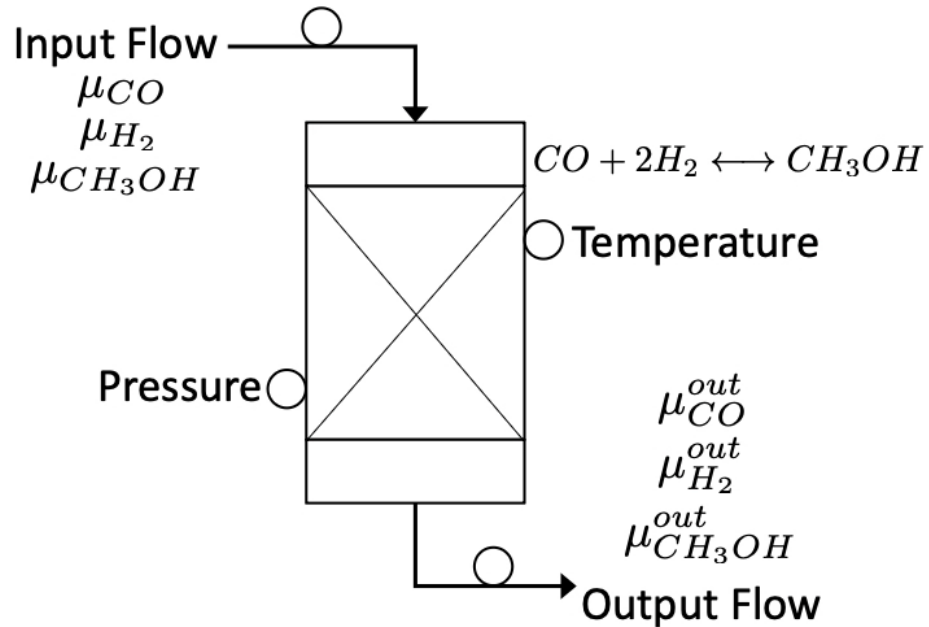
Revisit these questions –

1. What is known and unknown about this system? – sensor noise
2. What is the source of uncertainty?

Sensor measurement (location/sensitivity) - +/- 1% error margin

Importance of having sufficient data - allows us to analyze whether it is consistent or there are trends in the error/dependence on a specific variable.

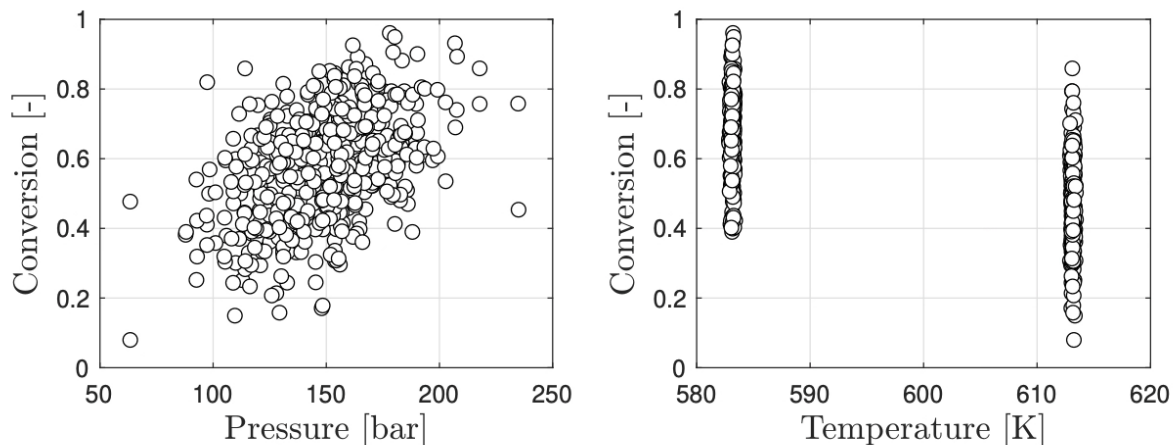
Another example – reactor unit



- System runs over sufficient period of time.
- Equilibrium achieved for reaction.
- **Two sensors – pressure and temperature.**
- P [50-250 bar] & T [583, 613K] – used to control the reaction.
- Can we determine how the P & T impact the conversion (fraction of reactant CO consumed in the reaction)?

$$\xi = 1 - \frac{\mu_{CO}^{out}}{\mu_{CO}}$$

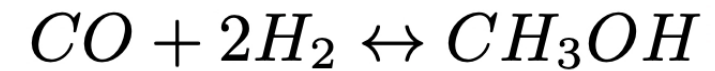
- We run experiments and measure – **input flowrate, temperature and pressure at the output.**



Data – P & T

- Fundamental conservation and thermodynamic laws – allow us to predict conversion from P & T.

$$\mu_k^{out} = \mu_k + \gamma_k \cdot \xi \cdot \mu_{CO}, k \in K$$



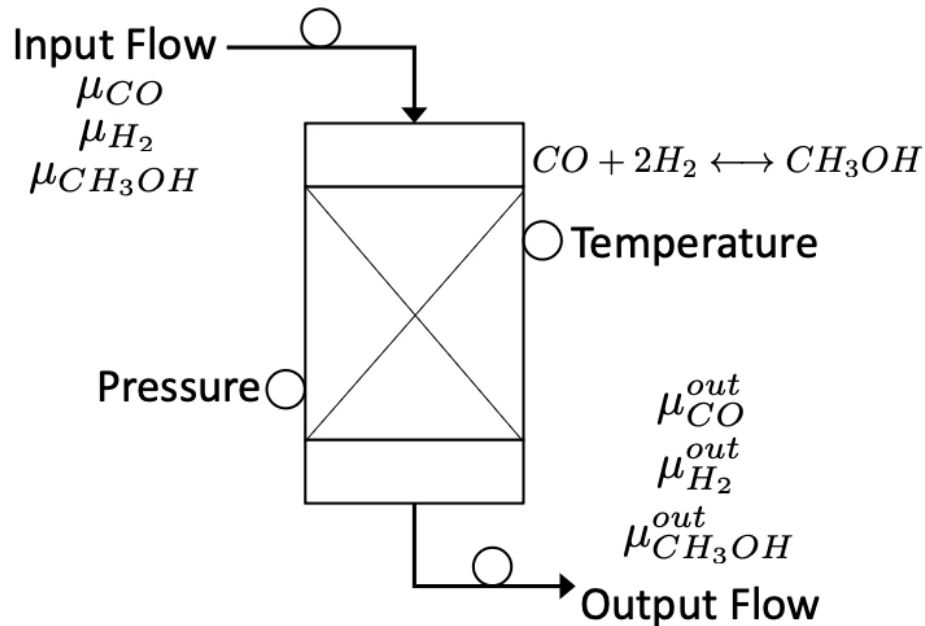
$$K_{eq} = \frac{(\mu_{CH_3OH} + \xi \cdot \mu_{CO})}{(\mu_{CO} - \xi \mu_{CO})(\mu_{H_2} - 2 \cdot \xi \cdot \mu_{CO})^2} \left(\frac{\mu_{out}}{P} \right)^2$$

What is known and what is unknown?

$$\frac{\partial \log K_{eq}}{\partial T} = -\frac{\Delta H_0}{R \cdot T}$$

- Van't Hoff equation tells us how the equilibrium is affected by temperature.
- Assuming $\Delta H_0 < 0$ (the reaction is exothermic), decreasing temperature will increase Keq.
- Increasing Keq – increases conversion.
- Temperature/Pressure profile understood from a mechanistic perspective!
- But variability?

Potential avenues for variability



- Equations are under ideal conditions – low pressure/high temperature.
- Uncertainty – non-ideal conditions?
- What is reactor is not well-mixed?
- What is sensor data is not accurate at these conditions?
- Location of sensor?

Having some prior knowledge of fundamental laws can drastically reduce the number of experiments needed to mitigate the uncertainty.
Then we can focus on the uncertain part – unexplained by the model.

Don't forget:

Inherent limitation of engineering practice: no matter how sophisticated our predictive models and sensing devices are, we will *a/ways* have a certain degree of uncertainty.

Characterizing both known and unknown aspects of a system is key.

Random Variables (RVs)

- RVs are basically used to capture the unknown or uncertain aspects.
- Example – water flow meter on a tap – fixed or changing?
- **Deterministic perspective –**
 - we ignore the presence of uncertainty and associated variability.
- Example – Room temperature – (causes of variability?)
- **Statistical perspective -**
 - Provide a range of possible values that capture the inherent variability/uncertainty of the value reported.
 - Data needed

Random Variables (RVs)

- Statistical vs deterministic view –

Deterministic Thinking: Temperature is 20°C (no uncertainty)

Statistical Thinking: Temperature is $20 \pm 0.5^{\circ}\text{C}$ (account for uncertainty)

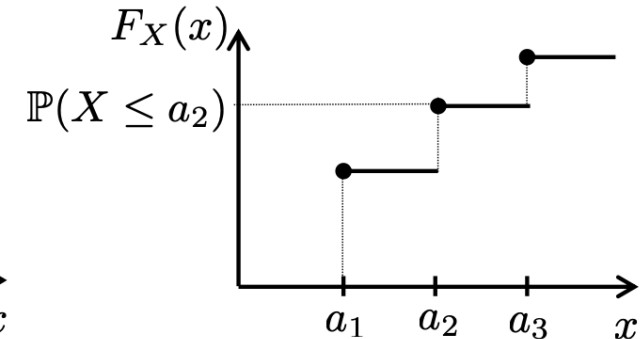
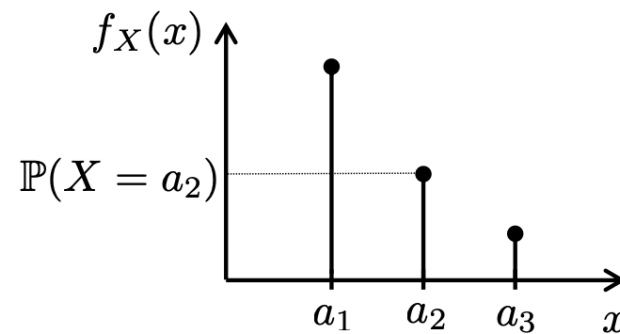
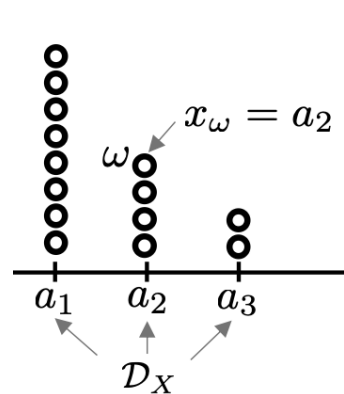
- RVs are used to model unknown or random behavior
- RVs do not have a known value and exhibits variability.

- Set of possible realizations $\omega \in \Omega$ with associated values $x_{\omega} \in \mathcal{D}_X$.
- Domain $\mathcal{D}_X$ under which realizations x_{ω} of X “live”.
- Probability measure $\mathbb{P} : \Omega \rightarrow [0, 1]$ that assigns probability to events.
- Cumulative density function (cdf) $F_X : \mathcal{D}_X \rightarrow [0, 1]$ that tells us $\mathbb{P}(X \leq t)$.
- Associated with cdf, there is a probability density function (pdf) $f_X : \mathcal{D}_X \rightarrow [0, \infty)$.

Random Variables (RVs)



- Think of ω as being a pinball that carries data x_ω
- When you drop a pinball, this will “fall” in different locations of the domain $\mathcal{D}_X$
- pdf/cdf tell us how “densely” pinballs accumulate in certain locations of the domain.
- Where in the domain the pinball falls is an event and probability tell us how likely are specific event.
- An event is a subdomain $\mathcal{A} \subseteq \mathcal{D}_X$ (location or set of locations in domain).
- Probabilities are related to how densely pinballs accumulate in parts of domain.

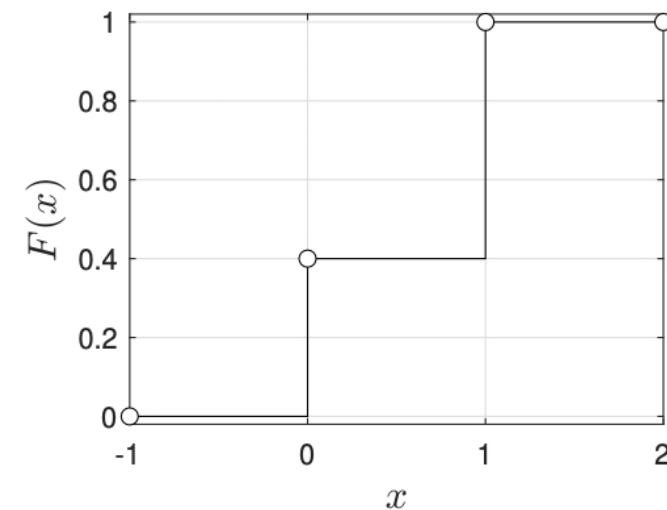
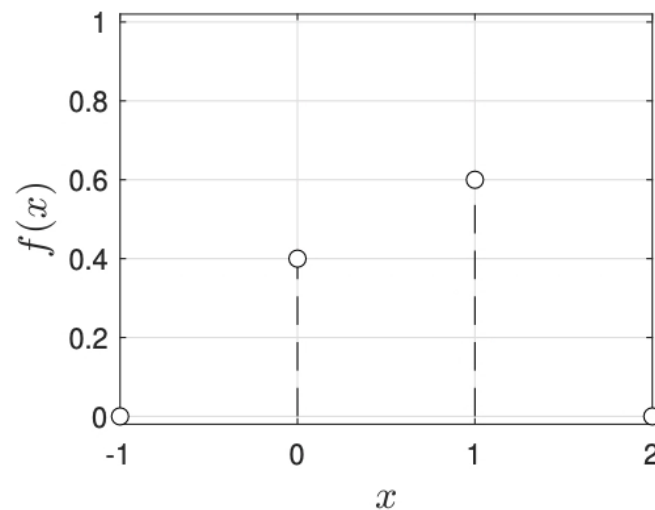
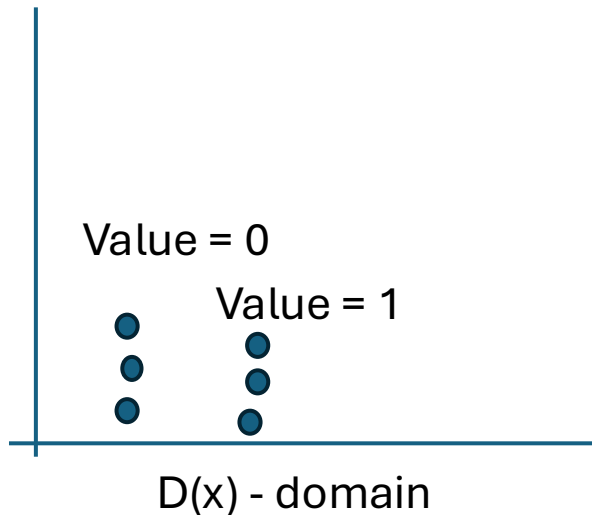


Random Variables (RVs)

Random variables (RVs) is a model that contains the following –

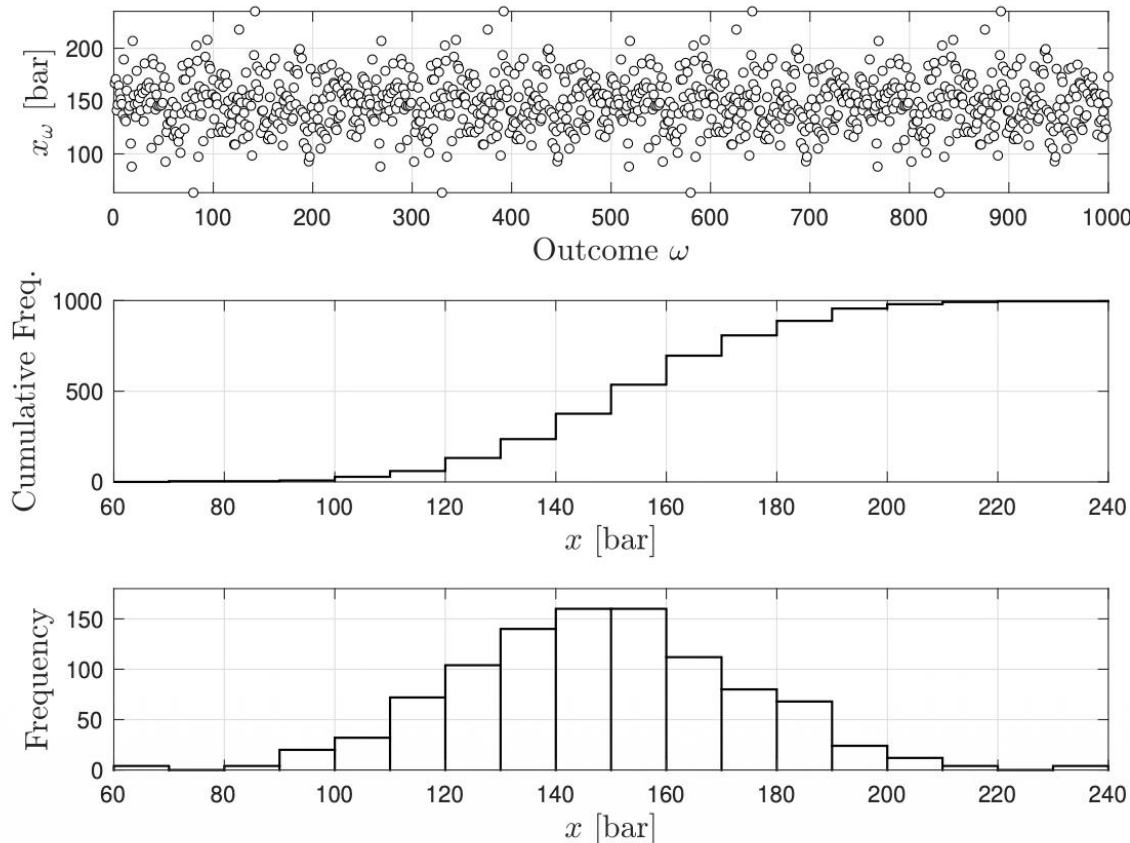
1. Set of realizations ($w \in Q$) with associated values x_w
2. Domain D_x under which values lie.
3. Probability measure - P
4. Probability density function PDF $f(x)$ – summarizes probabilities.
5. Cumulative density function CDF $F(x)$

X – discrete variable
Either 0 or 1 or 2 ...



Example – Gibbs' reactor --

- Assume *pressure* varies due to malfunction of flow control and model this as RV X
- Collect 1000 realizations x_ω (data); visualize as cumulative frequency and frequency.
- Cumulative frequency approximates cdf, frequency approximates pdf.

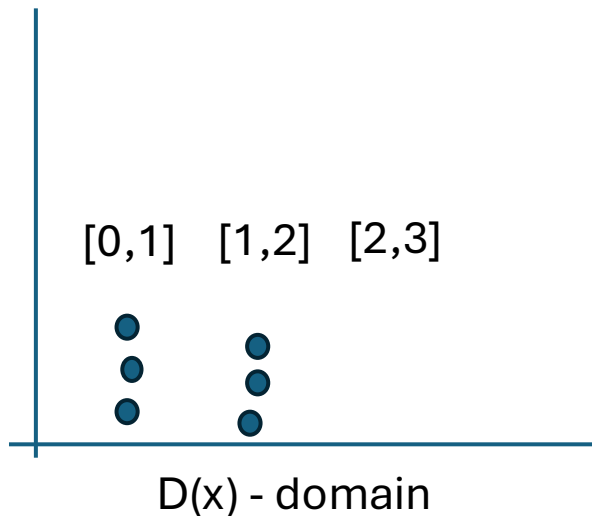


Random variables – univariate vs multivariate

- A *multivariate* RV $X = (X_1, X_2, \dots, X_n)$ has realizations that generate vector values $x_\omega = (x_{\omega,1}, x_{\omega,2}, \dots, x_{\omega,n}) \in \mathbb{R}^n$.
 - A *univariate* RV X is a multivariate with $n = 1$ and has realizations generate scalar values $x_\omega \in \mathbb{R}$.
-
- For a univariate RV, a pinball ω (observation) carries a single number x_ω (e.g., temperature).
 - For a multivariate RV, a pinball ω carries a set of numbers $x_\omega = (x_{\omega,1}, x_{\omega,2}, \dots, x_{\omega,n})$ (e.g., temperature, pressure, conversion)

Random Variables – RVs for continuous variables

- Similar analogy but for continuous variables –
- For ease of understanding we can partition the domain into bins –
- Probability measure allows us to understand how likely it is for the realizations to land in the bins and calculate probability of events.



- If we keep reducing the bin size - - > density function forms a continuous function.
- As bin size goes to 0, the probability of exact value breaks – $X = x$
- Instead, CDF makes more sense - - probability that value will be lower than a certain value as opposed to the exact value.

Amongst univariate, types of RVs – continuous & discrete

Discrete X has discrete domain $\mathcal{D}$ and cdf/pdf are discontinuous functions.

The pdf and cdf have the following properties:

$$\textcircled{1} \quad f(x) \geq 0, \quad x \in \mathcal{D}$$

$$\textcircled{2} \quad f(x) = \mathbb{P}(X = x), \quad x \in \mathcal{D}$$

$$\textcircled{3} \quad \sum_{x \in \mathcal{D}} f(x) = \sum_{x \in \mathcal{D}} \mathbb{P}(X = x) = 1$$

$$\textcircled{4} \quad \mathbb{P}(X \in \mathcal{A}) = \sum_{x \in \mathcal{A}} f(x), \quad \mathcal{A} \subseteq \mathcal{D}.$$

$$\textcircled{5} \quad F(t) = \mathbb{P}(X \leq t) = \sum_{x \in \mathcal{D} \mid x \leq t} f(x) \quad t \in \mathbb{R}.$$

PDFs of discrete RVs

A discrete RV is easy to handle computationally (simple summations and counting):

- Since $\mathcal{D}$ is discrete, we have that:

$$\begin{aligned}\mathbb{P}(X \leq t) &= F(t) \\ &= \sum_{x \in \mathcal{D} | x \leq t} f(x) \\ &= \sum_{x \in \mathcal{D}} \mathbf{1}[x \leq t] f(x).\end{aligned}$$

Here, we use the indicator function:

$$\mathbf{1}[x \leq t] = \begin{cases} 1 & \text{if } x \leq t \\ 0 & \text{if } x > t \end{cases}.$$

PDFs of discrete RVs

- Can compute pdf and cdf by *counting* how many times realizations x_ω of X take a certain value (or are below a certain value):

$$f(x) = \mathbb{P}(X = x) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \mathbf{1}[x_\omega = x]$$

$$F(t) = \mathbb{P}(X \leq t) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \mathbf{1}[x_\omega \leq t].$$

- We can thus determine pdf and cdf of discrete RV X directly from data x_ω .

Example – PDFs of discrete RVs

Consider a discrete RV with domain $\mathcal{D} = \{-1, 0, 1, 2\}$ and pdf:

$$f(x) = \begin{cases} 0 & \text{for } x = -1 \\ 0.4 & \text{for } x = 0 \\ 0.6 & \text{for } x = 1 \\ 0 & \text{for } x = 2 \end{cases}$$

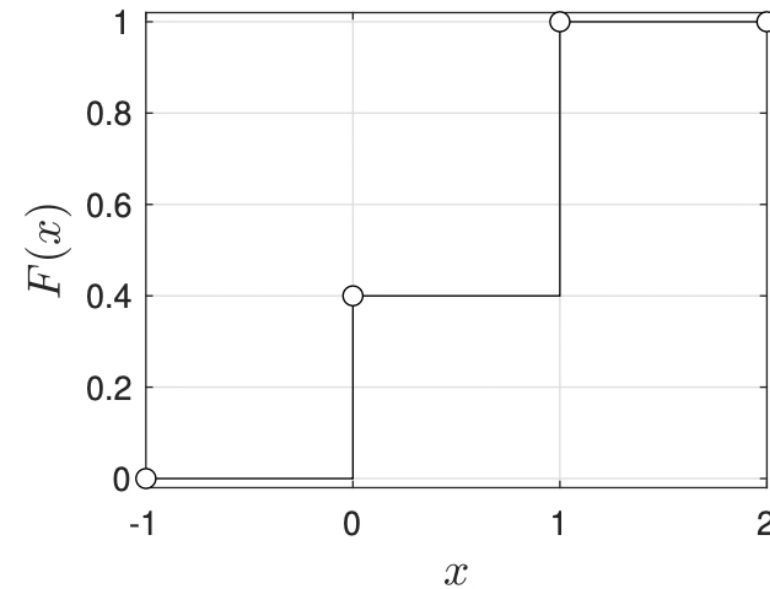
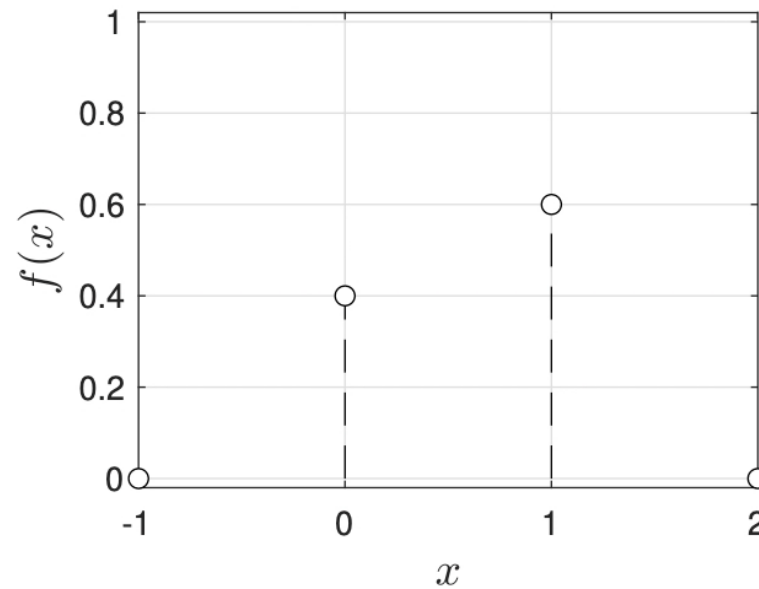
- Check if it follows PDF $\sum_{x \in \mathcal{D}} f(x) = 1$ and $f(x) \geq 0$ for all $x \in \mathcal{D}$.

Pdf can also be computed using the observations, for example:

$$f(0) = P(X = 0) = \frac{1}{10} \sum_{\omega \in \Omega} \mathbf{1}[x_{\omega} = 0] = 0.4$$

Example – PDFs of discrete RVs

Pdf and cdf of the RV are:



- `ch1_pdf_cdf_discrete`

PDFs of continuous RVs

Continuous X has continuous domain $\mathcal{D}$ and pdf/cdf are continuous functions.

The pdf and cdf have the following properties:

① $f(x) \geq 0, \quad x \in \mathcal{D}.$

② $f(x) = \frac{dF(x)}{dx}, \quad x \in \mathcal{D}.$

③ $\int_{x \in \mathcal{D}} f(x) dx = 1.$

④ $\mathbb{P}(X \in \mathcal{A}) = \int_{x \in \mathcal{A}} f(x) dx, \quad \mathcal{A} \subseteq \mathcal{D}.$

⑤ $F(t) = \int_{x \in \mathcal{D} | x \leq t} f(x) dx.$

Comparing discrete vs continuous RVs

① $f(x) \geq 0, \quad x \in \mathcal{D}$

② $f(x) = \mathbb{P}(X = x), \quad x \in \mathcal{D}$

③ $\sum_{x \in \mathcal{D}} f(x) = \sum_{x \in \mathcal{D}} \mathbb{P}(X = x) = 1$

④ $\mathbb{P}(X \in \mathcal{A}) = \sum_{x \in \mathcal{A}} f(x), \quad \mathcal{A} \subseteq \mathcal{D}.$

⑤ $F(t) = \mathbb{P}(X \leq t) = \sum_{x \in \mathcal{D} \mid x \leq t} f(x) \quad t \in \mathbb{R}.$

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⑤ $F(t) = \int_{x \in \mathcal{D} \mid x \leq t} f(x) dx.$

Comparing discrete and continuous RVs -

- **Continuous RVs need integrals** instead of summations.
- Computationally difficult to handle as compared to discrete RVs
- Finite difference approaches -

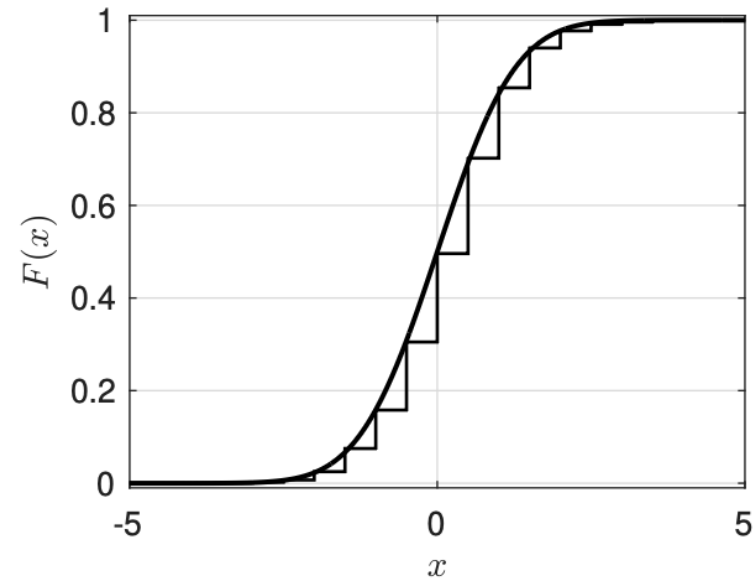
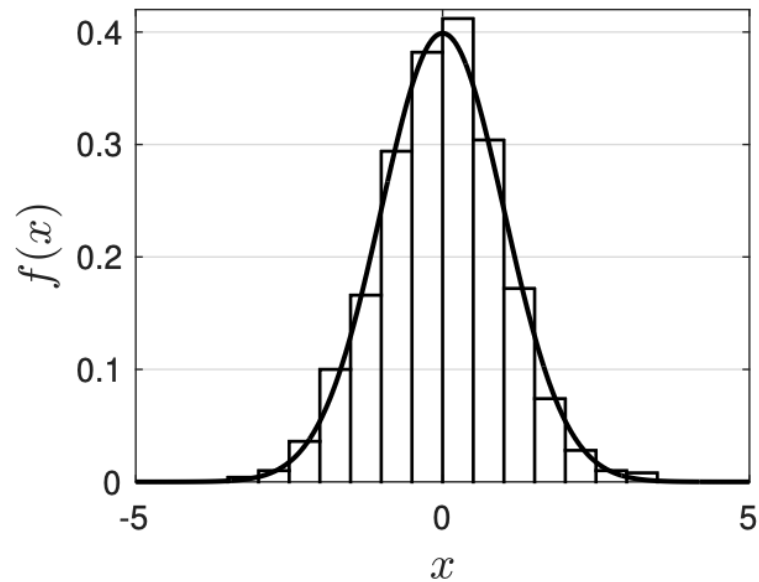
$$\begin{aligned}\mathbb{P}(X \leq t) &= \int_{x \in \mathcal{D} | x \leq t} f(x) dx \\ &= \int_{x \in \mathcal{D}} \mathbf{1}[x \leq t] f(x) dx.\end{aligned}$$

- Continuous RVs are often approximated using discretization. For instance:

$$f(x) \approx \frac{F(x + \Delta x) - F(x)}{\Delta x}.$$

In-class or HW – plot continuous variable – PDF and CDF -

Pdf and cdf of the RV are:



Example – Continuous RVs

- Consider continuous RV with domain $\mathcal{D} = (-\infty, \infty)$ and pdf of the form:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

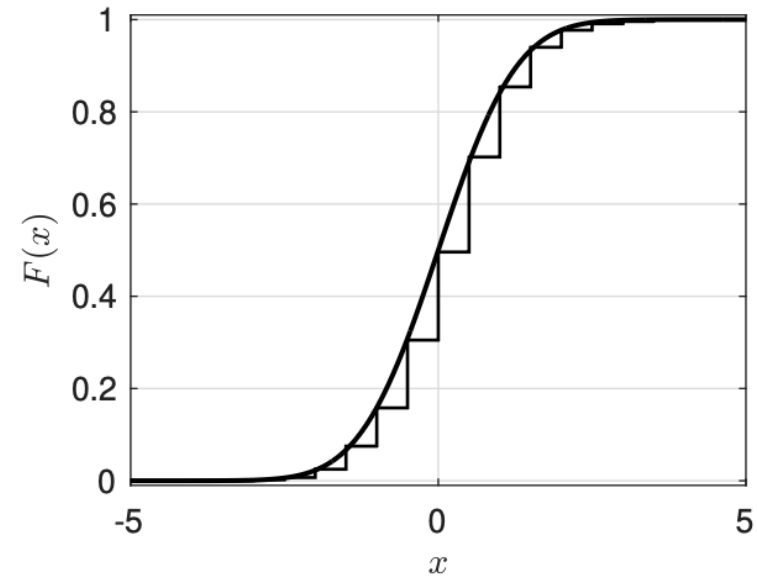
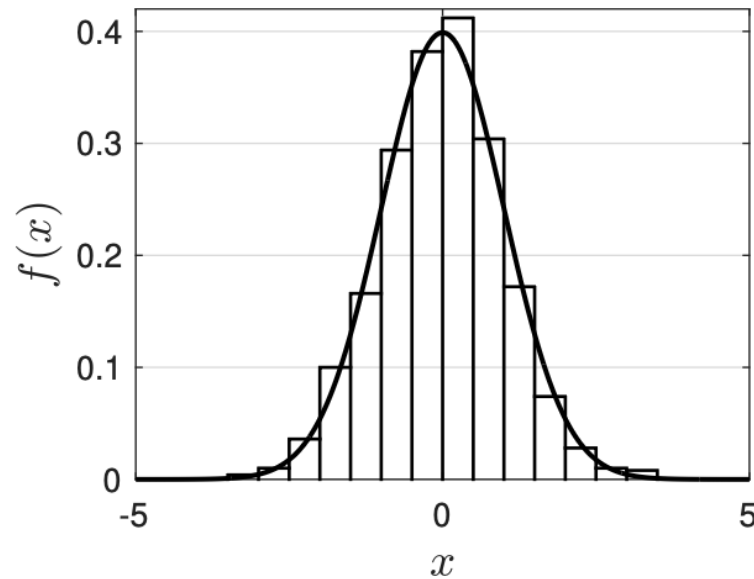
- It is easy to see that the pdf satisfies $f(x) \geq 0$ for any $x \in \mathcal{D}$ and one can show that $\int_{x \in \mathcal{D}} f(x) dx = 1$ (this is not that easy to show).
- The cdf associated with this pdf is given by:

$$\begin{aligned} F(t) &= \int_{x \leq t} f(x) dx = \int_{-\infty}^t f(x) dx \\ &= \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{t}{\sqrt{2}} \right) \right), \end{aligned}$$

- One can show $f(x) = dF(x)/dx$ and that cdf satisfies $0 \leq F(t) \leq 1$ for any $t \in \mathcal{D}$.

Example – continuous RVs

Pdf and cdf of the RV are:



- Calculated using approximate discretized bin size of 0.5

Data to Models --

DATA

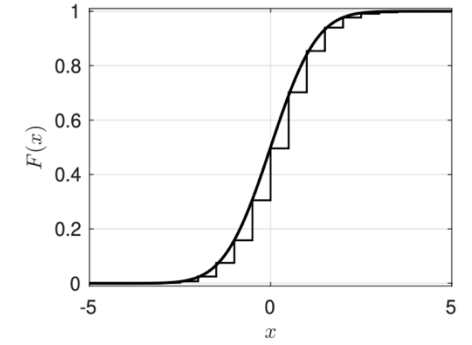
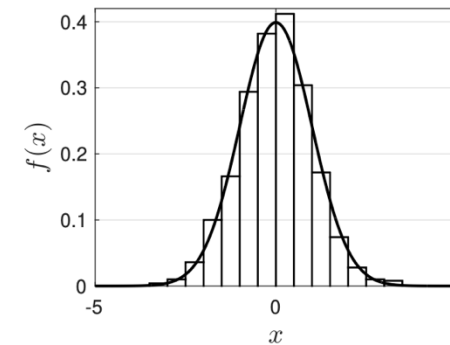
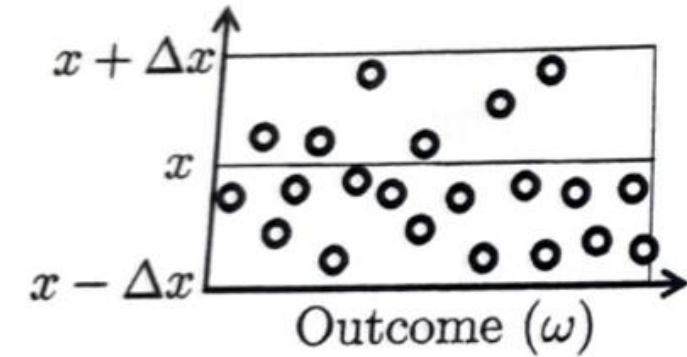
KNOWLEDGE

MODELS

- Practically, we obtain observations or data of the system under study to gain knowledge/understanding
- Knowledge - put in form of models – that can be used for proper predictions and decisions.
- Types of models –
 - 1. **Structural models** – characterization of structural dependencies between variables – mechanistic or empirical.
 - 2. **Random variable models or uncertainty-based** characterization of behavior that cannot be explained by structural models.

Data – to – Models

- Structural models – well known.
- **Random variable models** -
 - To construct one – what we do we need?
 - Observations $x_\omega, \omega \in \mathcal{S}$.



Two main questions –

1. What type of RV do the observations follow?
2. How to obtain PDF and CFD for a model from available data?

Data – to – Models

- Goal is to use available data x_ω to postulate a *theoretical* RV model X .
- RV X is a *model* of a random phenomenon that generates the observed data.
- Many models are available to capture diverse phenomena seen in real life.
- A widely used model is that of a Gaussian RV.
- Model assumes that X is continuous, has domain $\mathcal{D} = (-\infty, \infty)$, and has pdf:

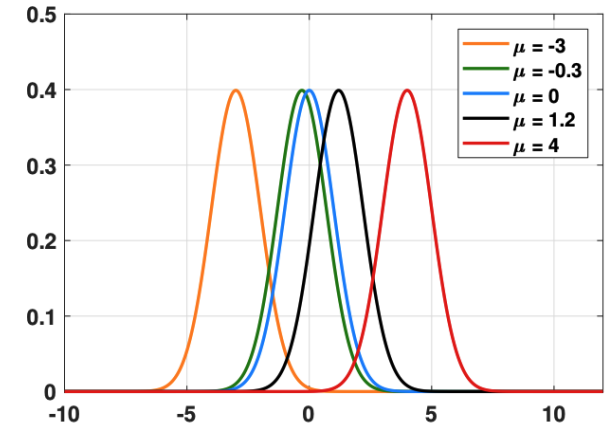
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathcal{D},$$

where $\mu, \sigma \in \mathbb{R}_+$ are parameters of the model.

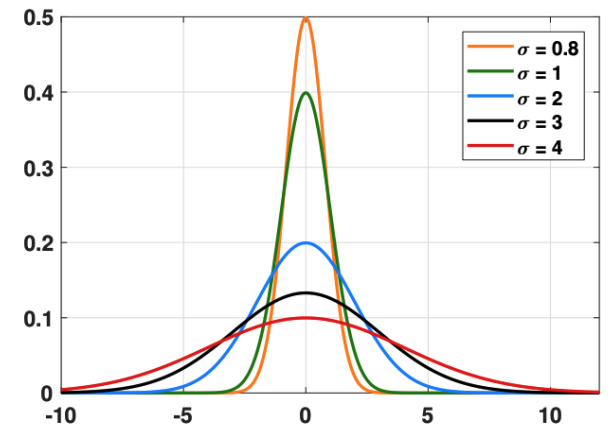
- Parameters define behavior of the RV and can be “tuned” to match our data.

The cdf associated to the Gaussian model is:

$$F(x) = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{(x-\mu)/\sigma}{\sqrt{2}} \right) \right), \quad x \in \mathcal{D},$$



μ changes, $\sigma = 1$



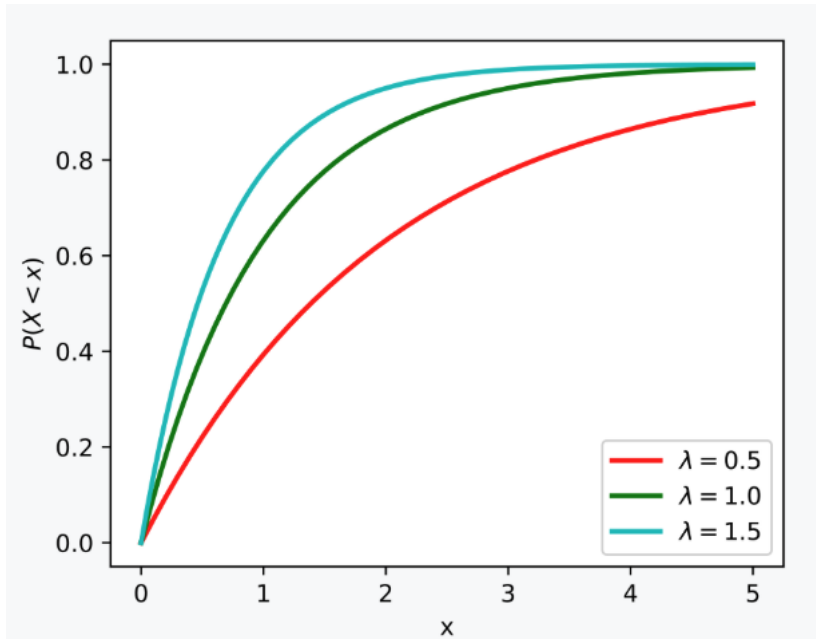
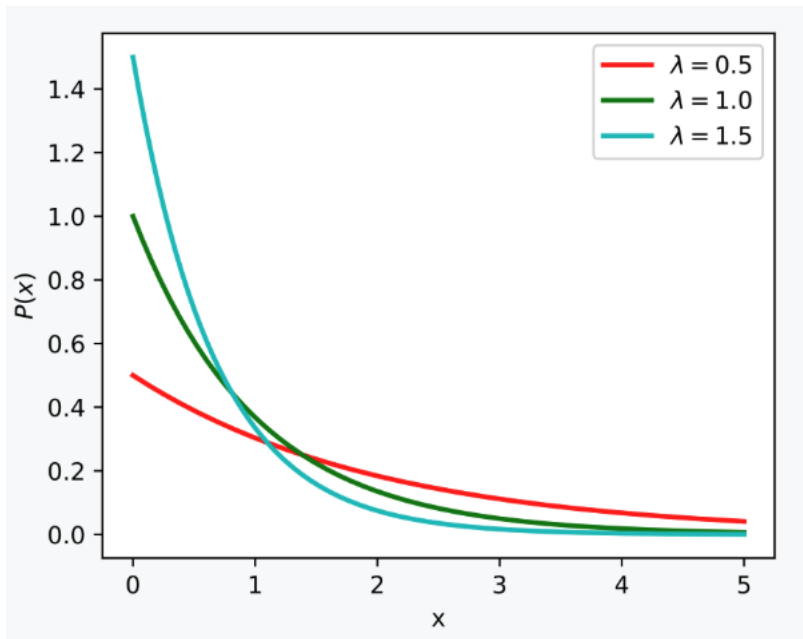
$\mu = 0$, σ changes

Data – To – Models

- **Gaussian RV – continuous or discrete?**
- **Exponential RVs. -**
 - Assumes X to be continuous with a domain $D=[0,\infty)$
 - PDF and CDFs included below

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

$$F(x; \lambda) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

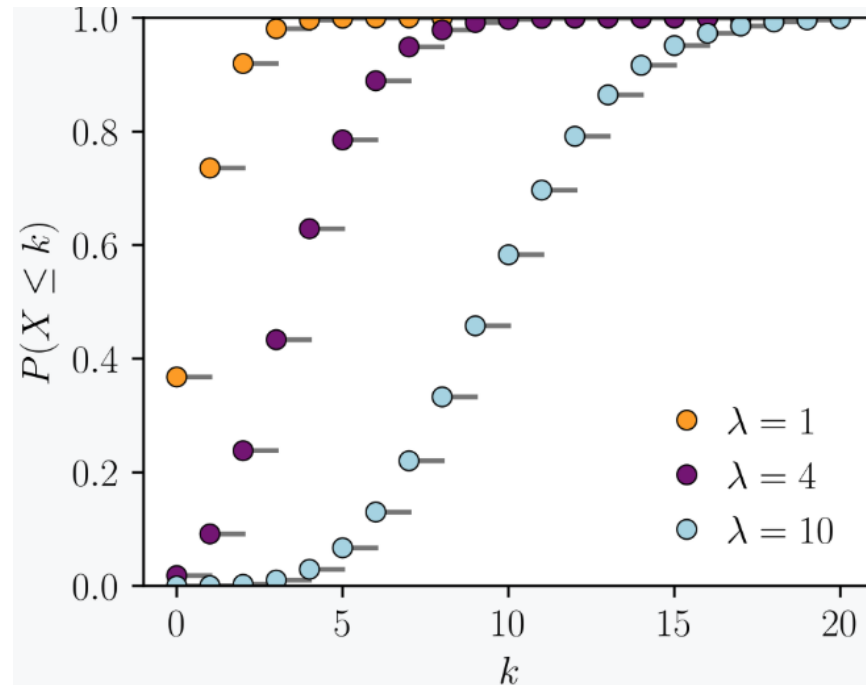
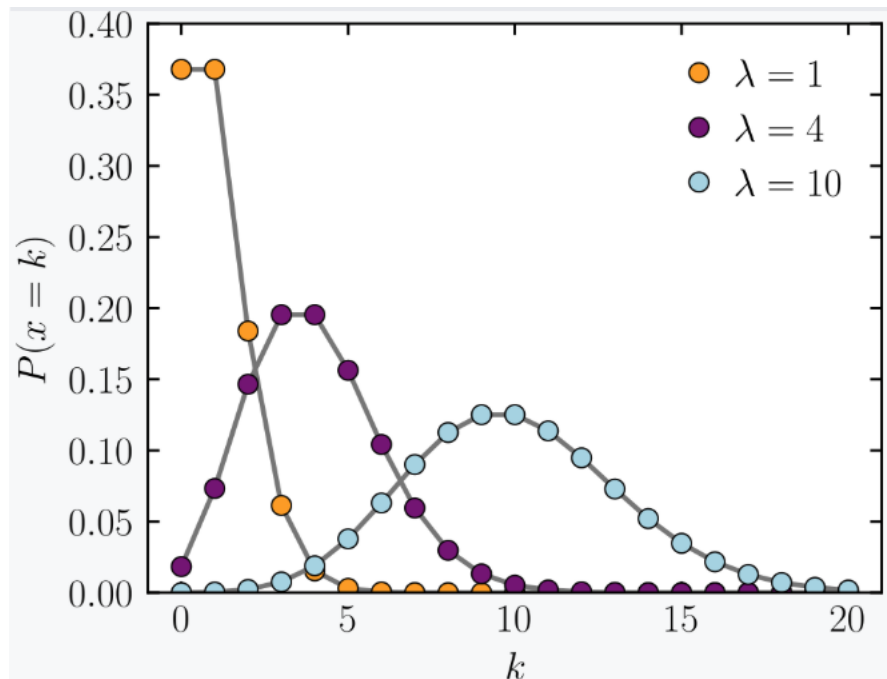


Data – To – Models

$$f(k; \lambda) = \Pr(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

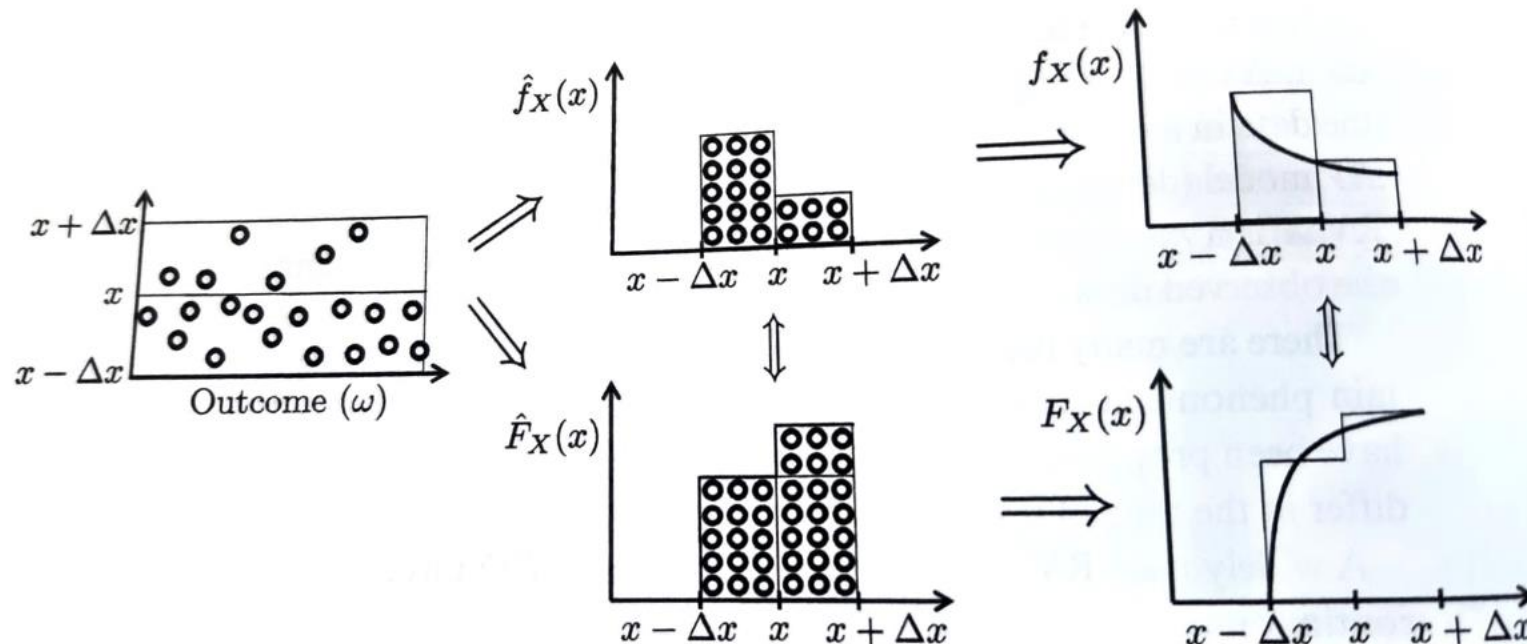
- **Poisson RVs. -**

- Assumes X to be discrete with a domain $D = \{0, 1, 2, \dots, \infty\}$
- PDF and CDFs included below with k denoting the number of occurrences.

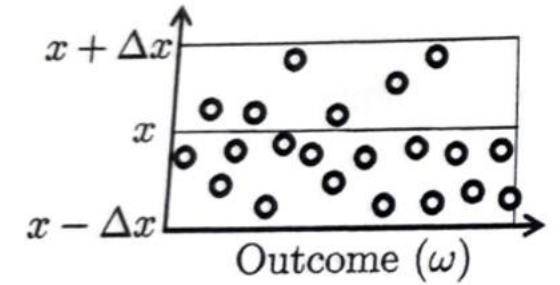


Data – To – Models

- Our GOAL – to investigate if the available observations follow the PDF and CDF of a theoretical model.
- Start with data collection,
- Construct **empirical approximations** of PDF and CDF – data-driven or sample-based.
- Discuss empirical vs population



Empirical approximations for PDF and CFD



- 1. Empirical Domain D – initial indication of continuous or discrete variable.
- For continuous, empirical PDF = total number of observations x_w that lie in the interval $[x, x + \Delta x]$ divided by the bin size and number of observations.

$$\hat{f}(x) = \frac{P(x \leq X \leq x + \Delta x)}{\Delta x}$$

$$\hat{f}(x) = \frac{\frac{1}{S} (\sum_{w \in S} [x \leq x_w \leq x + \Delta x])}{\Delta x}$$

Empirical approximations for PDFs & CDFs

- For discrete, empirical pdf =

$$\hat{f}(x) = \frac{1}{N} \sum_{x_w \in S} [x_w = x]$$

- Empirical PDF = Total no. of observations where $x_w = x$ / no. of observations
- Empirical CDF =

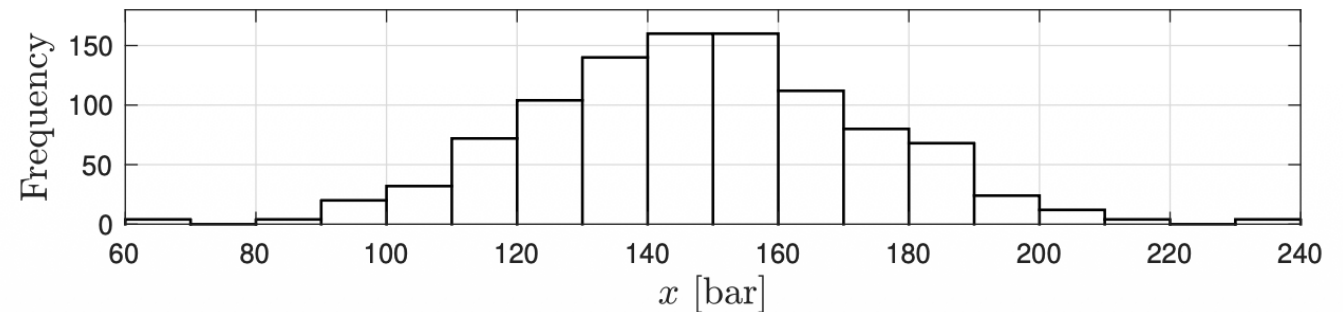
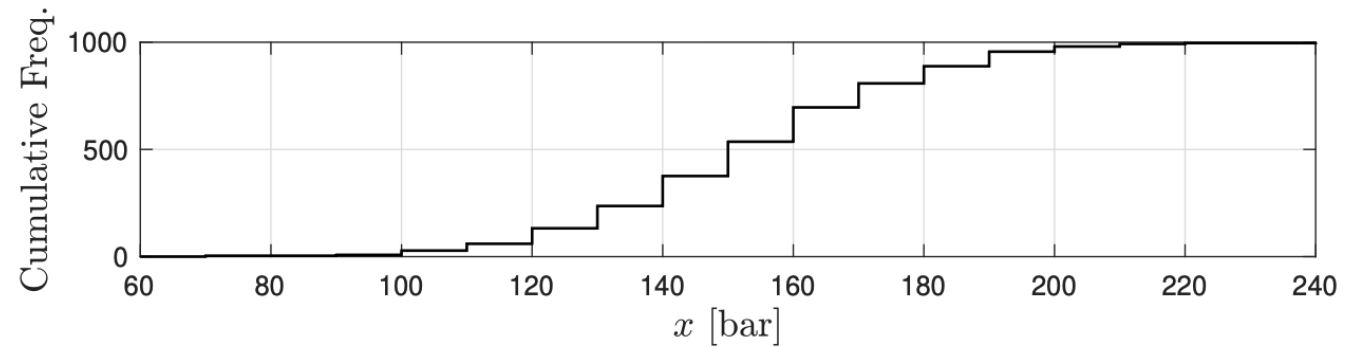
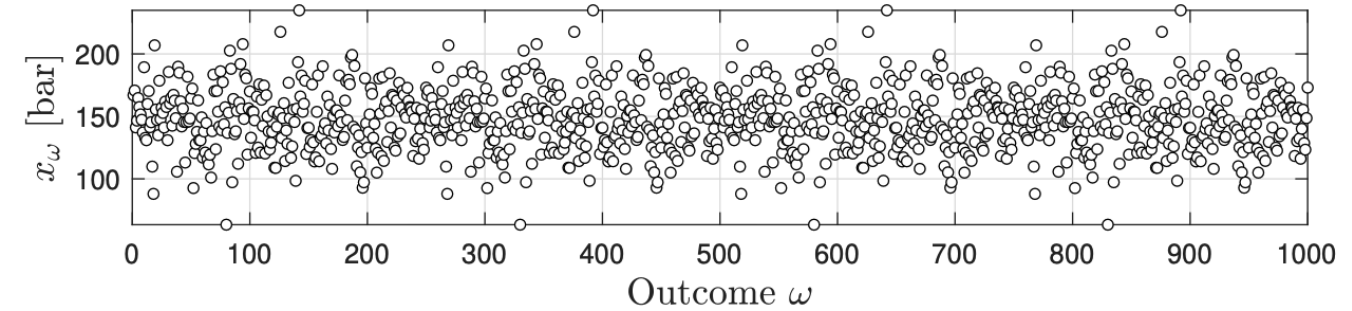
$$\hat{f}(x) = \frac{1}{N} \sum_{x_w \in S} [x_w \leq x]$$

Discrete PDF/CDF – much easier as the domain does not need to be discretized

Goal of empirical CFD/PDF =
Can we predict the outcomes that have not been observed so far?

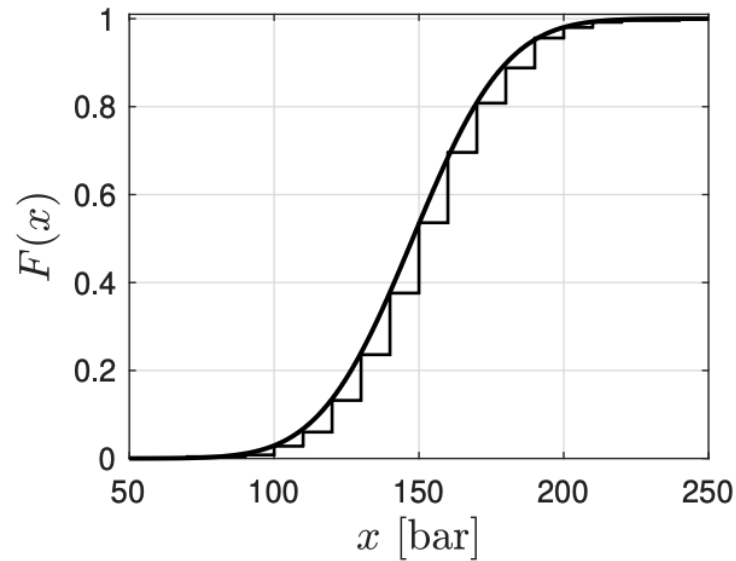
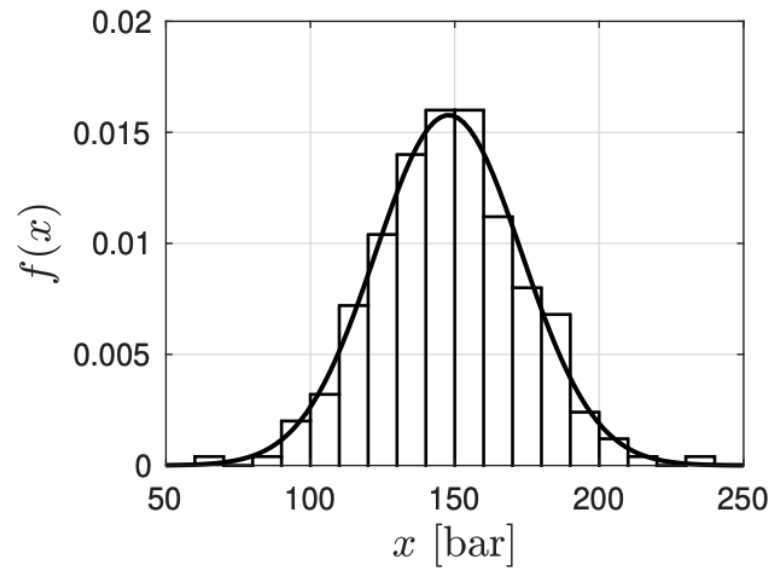
In-class example – Data from Gibbs reactor

- ch1_gibbs_example.m



In-class example – Gibbs reactor

- Empirical PDF and CDF



- Fitting them to a RV model – Gaussian → **Conclusion: our random phenomena behaves as a gaussian RV.**

Why do all this to find a theoretical model that fits the random data?

1. Model indicates that the data is not arbitrary but has a trend – extracting trends gives us **knowledge about the data**.
2. We can now use our RV model to make predictions about outcomes that you have not seen before. Eg. How likely is it that the reactor takes an extreme pressure that has not been seen before? – equipment safety and failure analysis.
3. Ability to predict probability to different outcomes.
4. RV models – **predicts the probability of outcomes, not the outcome itself..** Do you see the difference from structural models?

Summarizing statistics

- Statistics can be used as measures of risk.
- Popular measures – expected mean and variance --

Discrete RV –

Expected mean

$$\mathbb{E}_x = \sum_{x \in D} x * f(x)$$

Variance and standard deviation

$$\mathbb{V}_x = \sum_{x \in D} (x - \mathbb{E}_x)^2 * f(x)$$

Continuous RV –

Expected mean –

$$\mathbb{E}_x = \int_{x \in D} x * f(x)$$

Variance and standard deviation - $\mathbb{SD}_x = \sqrt{\mathbb{V}_x}$

$$\mathbb{V}_x = \int_{x \in D} (x - \mathbb{E}_x)^2 * f(x)$$

What if we do not have $f(x)$ – theoretical PDF?

- Data-driven estimates
- Empirical estimates for mean and variance –

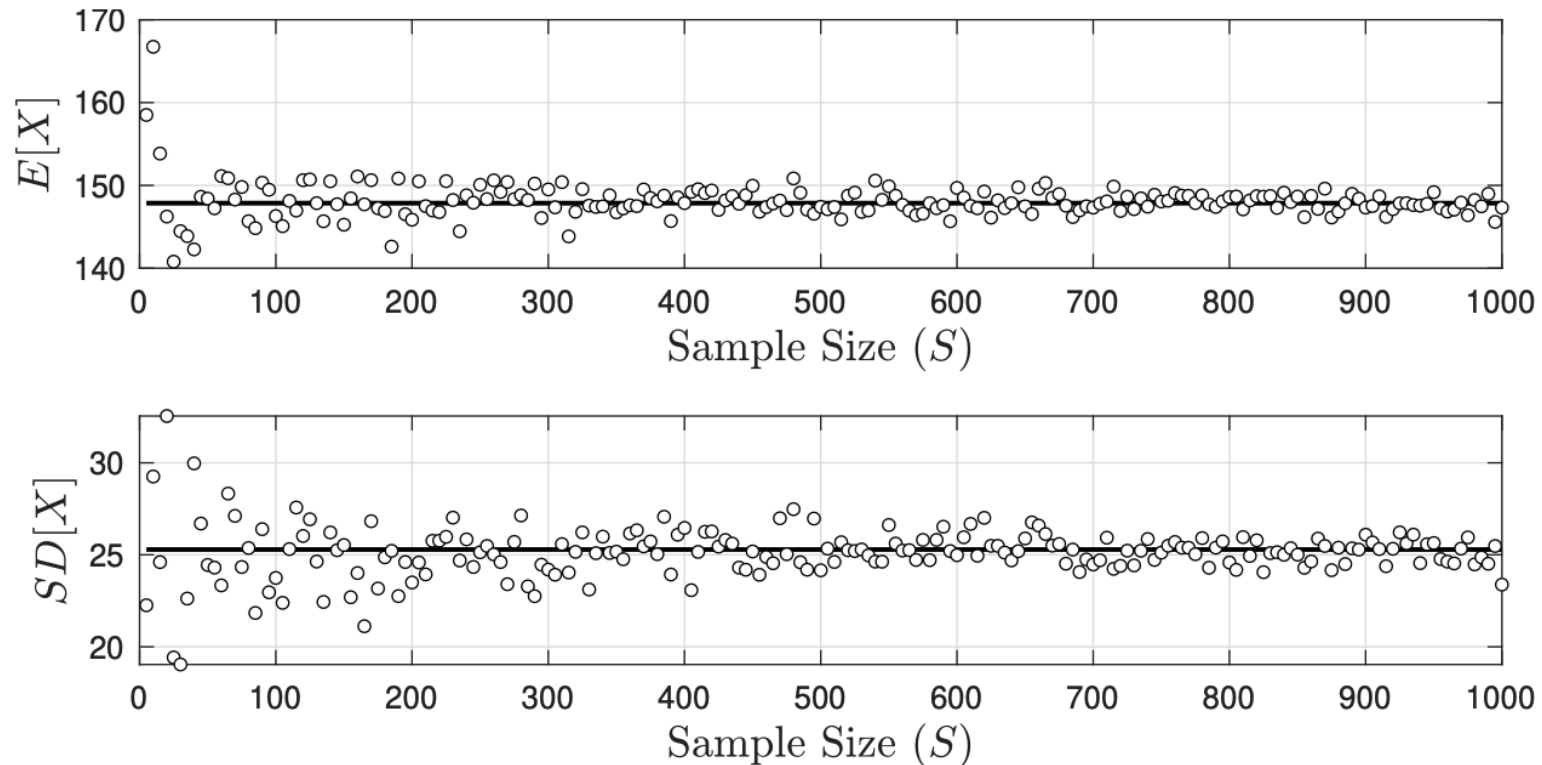
$$\hat{\mathbb{E}}_x = \frac{1}{S} \sum_{x_w \in S} x_w$$

$$\mathbb{V}_x = \frac{1}{S} \sum_{x_w \in S} (x_w - \hat{\mathbb{E}}_x)^2$$

- Useful descriptors of datasets.
- As data increases, the empirical estimates come closer to the theoretical statistics.

In-class example – ch1_gibbs_example

- Calculate mean and standard deviation

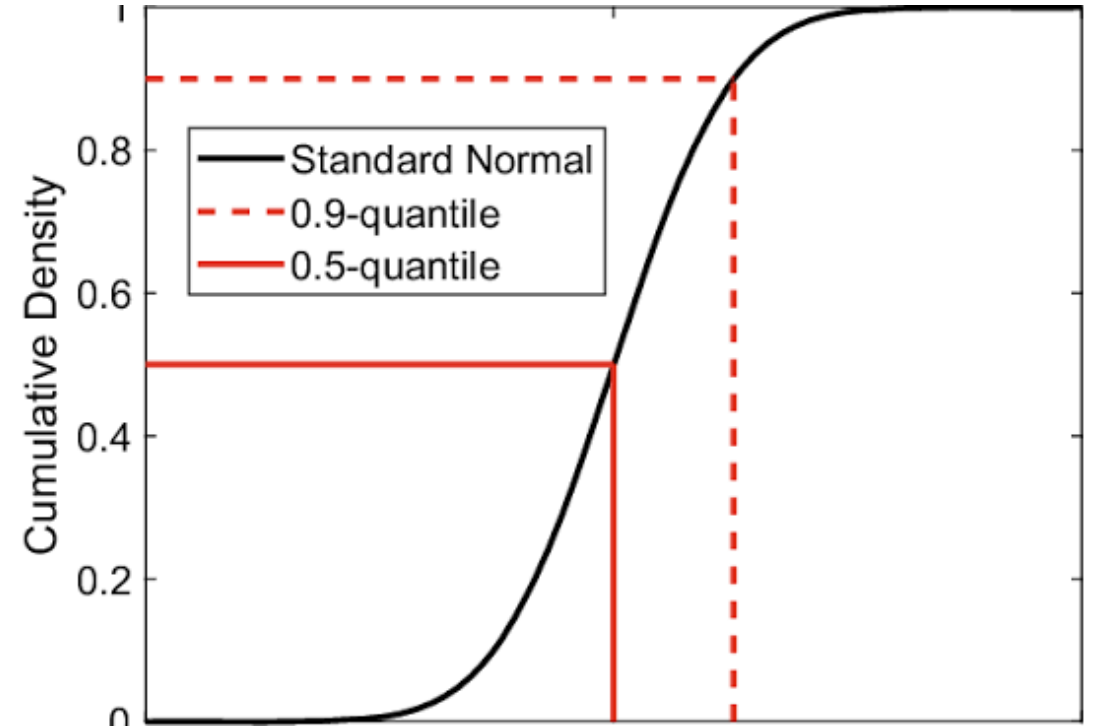


Quantiles – statistics

- Quantile is the inverse function of the CDF –

$$\mathbb{Q}_X(\alpha) = F_X^{-1}(\alpha)$$

- Quantile - tells you that a specific fraction or percentage of the data falls below that point.



$\mathbb{Q}_X(0.5)$ is the *center value* of X (a.k.a. the median and denoted as $\mathbb{M}_X$)

$\mathbb{Q}_X(1) = \max_{x \in \mathcal{D}_X} x$ is the maximum value of X

$\mathbb{Q}_X(0) = \min_{x \in \mathcal{D}_X} x$ is the minimum value of X

Empirical CDF to calculate
Empirical Quantile

Mode and moment - Statistics

- Mode of an RV is the outcome of maximum probability.

$$\text{MO}_X \in \operatorname{argmax}_{x \in \mathcal{D}_X} f_X(x).$$

- For example, for a Gaussian RV, mode = mean.
- However, we can have RVs with multimodal distribution instead of unimodal (one maximum point) – multimodal RVs

- Moments –

- Raw moment -

$$\text{RMO}_k[X] = \mathbb{E}[X^k], \quad k = 1, 2, 3, 4 \dots,$$

- Central moment –

- Standardized moment - $\text{CMO}_k[X] = \mathbb{E}[(X - \mathbb{E}[X])^k], \quad k = 1, 2, 3, 4, \dots,$

$$\text{SMO}_k[X] = \frac{\mathbb{E}[(X - \mathbb{E}[X])^k]}{\text{SD}[X]^k}, \quad k = 1, 2, 3, 4, \dots,$$

- Q. what is the 1st Raw Moment? 2nd Raw moment?
 - $E[X^k] = E[X]$
 - $E[X^2] = \mu^2 + \sigma^2$
- Central moment – shift the data by subtracting the mean, isolates the shape of the distribution from its location.
- Q. What is 1st central moment of a gaussian RV?

$$\text{RMO}_k[X] = E[X^k], \quad k = 1, 2, 3, 4 \dots,$$

$$\text{CMO}_k[X] = E[(X - E[X])^k], \quad k = 1, 2, 3, 4, \dots,$$

$$\text{SMO}_k[X] = \frac{E[(X - E[X])^k]}{\text{SD}[X]^k}, \quad k = 1, 2, 3, 4, \dots,$$

- SMO – remove mean and SD both – dimensionless numbers to evaluate the shape of the distribution regardless its units.

For a Gaussian random variable -

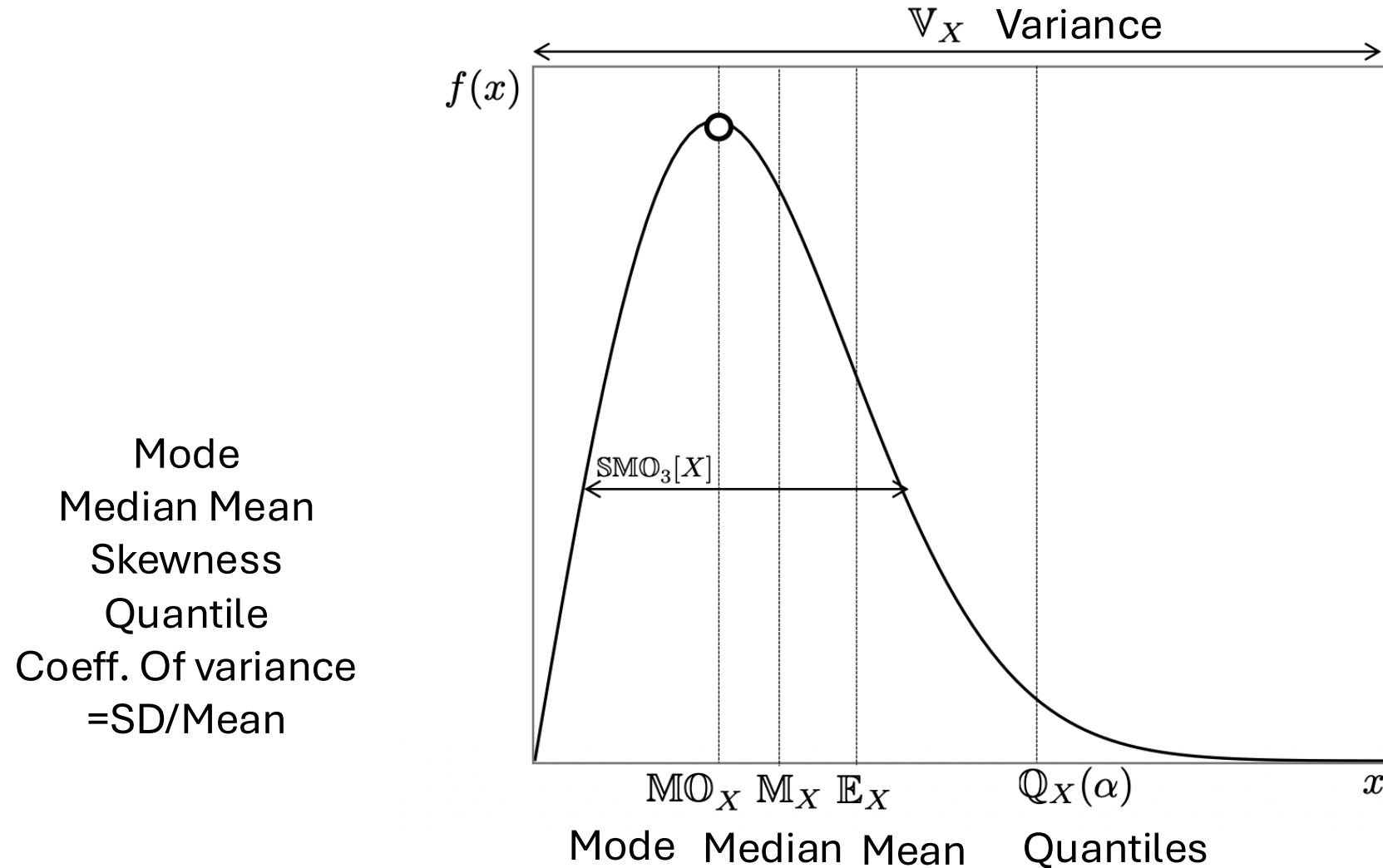
Order (n)	Raw Moment ($E[X^n]$)	Central Moment ($E[(X - \mu)^n]$)	Standardized Moment ($E[Z^n]$)	Common Statistical Name
1	μ	0	0	Mean (1st Raw)
2	$\mu^2 + \sigma^2$	σ^2	1	Variance (2nd Central)
3	$\mu^3 + 3\mu\sigma^2$	0	0	Skewness (3rd Standardized)
4	$\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$	$3\sigma^4$	3	Kurtosis (4th Standardized)

Skewness – measure of symmetry of the PDF, >0: more weight on right side)

Kurtosis – measure of the nature of tails of the PDF

Excess kurtosis – metric that tells you how much the tails vary from that of a gaussian distribution

Features of PDF



Back to Gibbs_reactor example -

- Quantiles –
- Mode and moments

Where are these statistics useful?

- Event probability -

$$\mathbb{P}(a1 \leq x \leq a2) = \frac{1}{S} \sum_{x \in D} 1[a1 \leq x \leq a2]$$

If you have data

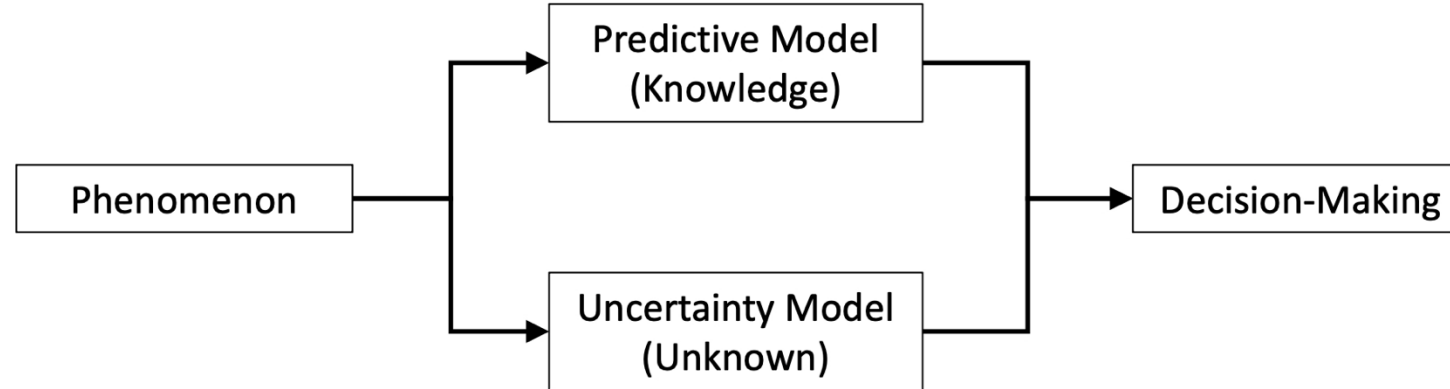
$$\mathbb{P}(a1 \leq x \leq a2) = \int_{x \in [a1, a2]} f(x)$$

If you have pdf $f(x)$, and can process integrals

Example – **decision making**

1. Determine the **probability** that the conversion is between $a1$ and $a2$
2. Measures of risk – eg. Compute the probability that an investment yields positive yield or the probability that a reactor fails.

Structural models



- Structural models differ from RV models where they aim to extract predictable behavior from data
- Main tools for developing structural models from data – experimental design and parameter estimation. $Y = f(x, \emptyset) + \epsilon$
- Here, f is the model prediction with $(\emptyset)$ being the parameter estimation
- Question – how to best select inputs that can maximize information gain – **DOE**
- Using information to optimize performance – **decision making**.

From Models to decisions

- We have a characterization of a **random variable (X)**
 - With a structural model that captures how this random input (u) and a deterministic input (x) affect Y.
 - $Y = f(x, u)$
 - Can we manipulate Y in desirable ways?
- Uncertainty propagates in complex ways – one variable introducing uncertainty in another. (e.g., Sensor error – impacting conversion)
 - Uncertainty propagation can be mitigated by design/control decisions.

Uncertainty Propagation & mitigation

- Consider propagation of X through system $\phi(X,u)$:

$$Y = \phi(X,u)$$

- where u is a mitigating action (decision – we decide its value) and Y is the system output.
- We make the following observations:
- Output $Y = RV$ if the input X is an RV .
- Nature of Y (its cdf, pdf, and domain) depends on system ϕ and u . Some systems magnify uncertainty and variability while others might damp it.
- Can use action (u) to mitigate/manipulate uncertainty of Y .

Uncertainty propagation

- For example – Consider propagation of uncertainty of wind power –
- The uncertain inputs X introduces uncertainty in the operating cost Y .
- However, the grid has available variables (u – eg. Generation from other plants) that can be used to counteract the wind power uncertainty and mitigate the uncertainty in Y .

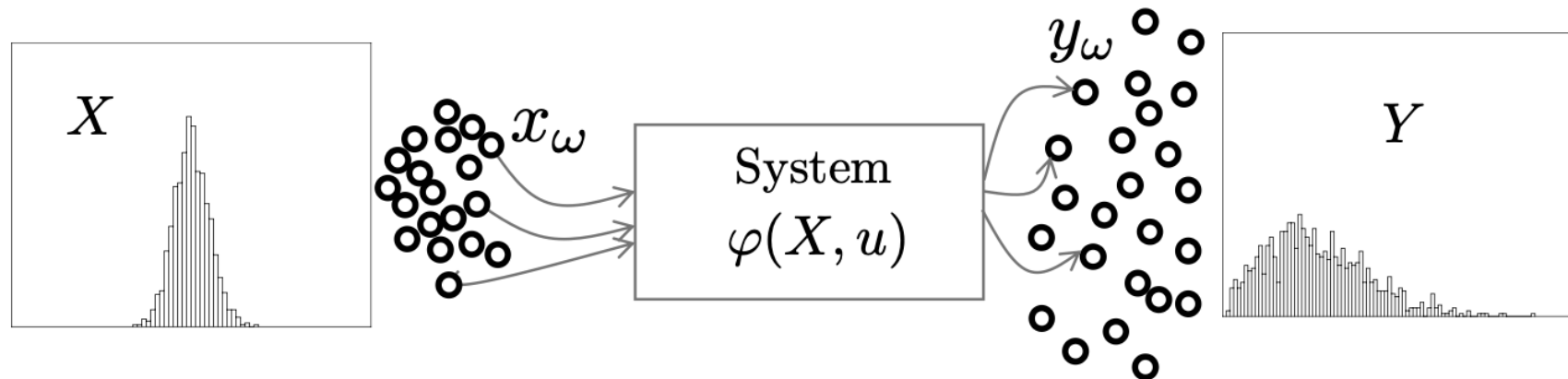


Figure: Illustration of propagation of a random variable X through a system. The propagation results in a random output Y that has different characteristics (e.g., higher variability/uncertainty).

Decision making amidst uncertainty

- To make decisions, we need to characterize the uncertainty.
- Having data x_ω , $\omega \in \mathcal{S}$ and a system model ϕ , we can characterize **cdf, pdf, domain, and summarizing statistics of Y** using the following simulation procedure:

$$y_\omega = \varphi(x_\omega, u), \omega \in \mathcal{S}$$

Known as Monte Carlo (MC) simulations – estimate diverse quantities of interest for RVs.

- Sample mean:

$$\hat{\mathbb{E}}_Y = \frac{1}{S} \sum_{\omega \in \mathcal{S}} y_\omega = \frac{1}{S} \sum_{\omega \in \mathcal{S}} \varphi(x_\omega, u)$$

- Sample variance:

$$\hat{\mathbb{V}}_Y = \frac{1}{S} \sum_{\omega \in \mathcal{S}} (y_\omega - \hat{\mathbb{E}}_Y)^2$$

- Empirical cdf:

$$\hat{F}_Y(y) = \frac{1}{S} \sum_{\omega \in \mathcal{S}} \mathbf{1}[y_\omega \leq y]$$

Estimates are good approximations of the actual expected mean and variance

MC

- Repeated random sampling -
- Dataset vs population (integral)
- **Overcomes integration of a complex function.**
- MC – highly versatile: used to understand how uncertainty propagates through a complex system, sensitivity analysis or numerical integration.

- Sample mean:

$$\hat{\mathbb{E}}_Y = \frac{1}{S} \sum_{\omega \in \mathcal{S}} y_\omega = \frac{1}{S} \sum_{\omega \in \mathcal{S}} \varphi(x_\omega, u)$$

- Sample variance:

$$\hat{\mathbb{V}}_Y = \frac{1}{S} \sum_{\omega \in \mathcal{S}} (y_\omega - \hat{\mathbb{E}}_Y)^2$$

- Empirical cdf:

$$\hat{F}_Y(y) = \frac{1}{S} \sum_{\omega \in \mathcal{S}} \mathbf{1}[y_\omega \leq y]$$

Domain (D)

$$\mathbb{E}_x = \int_{x \in D} x * f(x)$$

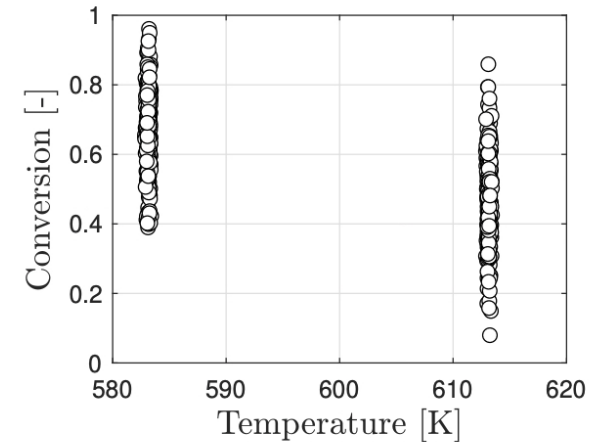
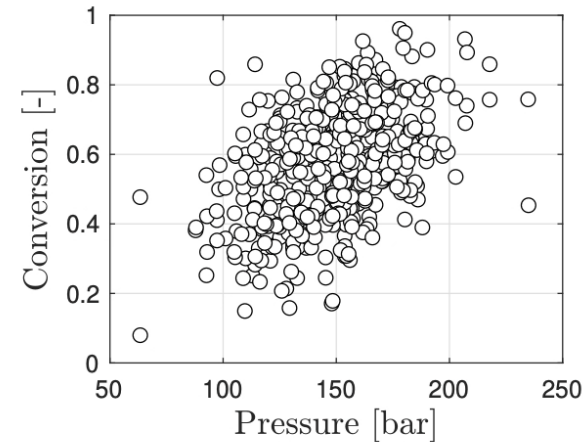
$$\mathbb{V}_x = \int_{x \in D} (x - \mathbb{E}_x)^2 * f(x)$$

Example – Uncertainty in gibbs reactor

- **Can we understand how the uncertainty in pressure generates uncertainty in the conversion -**

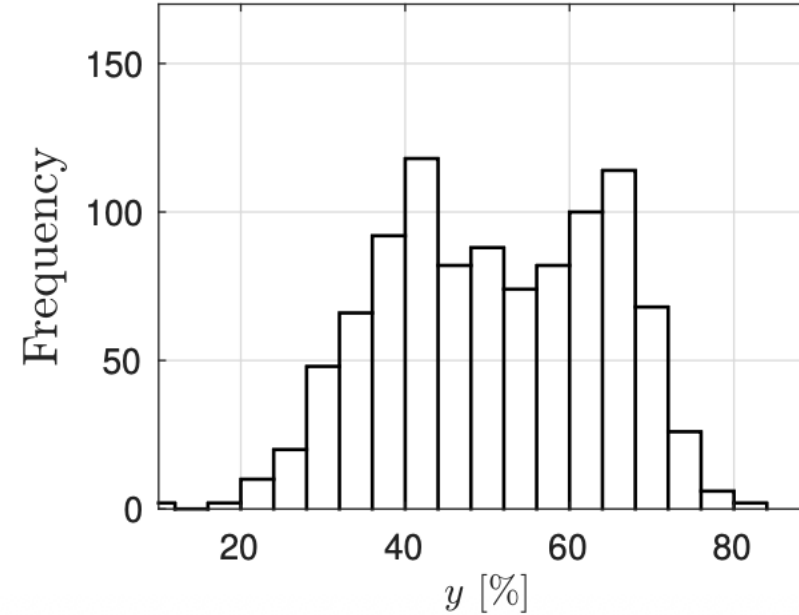
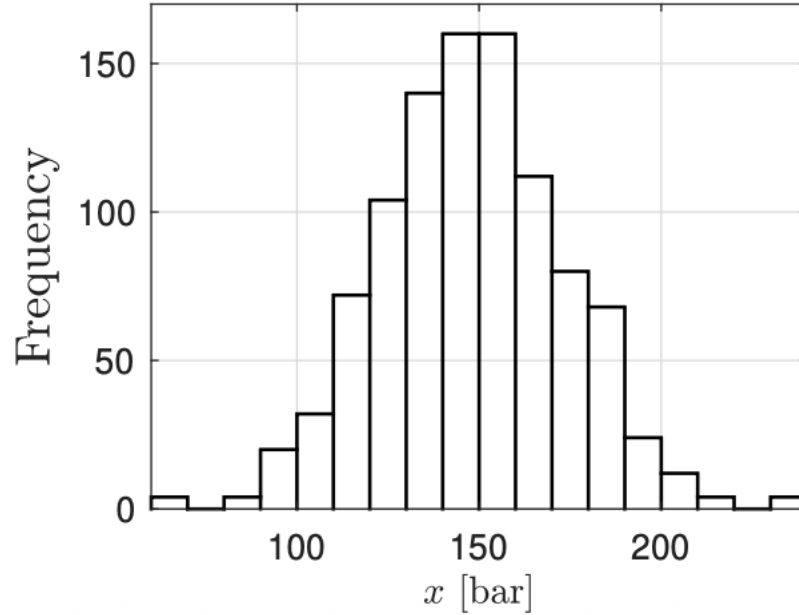
$$Y = \phi(X, u)$$

-
- X: pressure (RV)
- Y: conversion (RV)
- u: decision or control variable – temperature



- MC to obtain empirical PDF and CDFs, assuming the reactor is operated at high temperature.

Example- Gibbs reactor



- Unimodal behavior in X and bimodal behavior in Y
- Indicates **strong non-linear behavior between X and Y** – pressure and conversion.

Uncertainty to decision making --

- We would now like to find a **decision u** that **manipulates $Y(u) = \phi(X, u)$** in a desirable way (e.g., minimizing uncertainty/variance).
- Consider the questions:

If we have a couple of competing decisions u and u' giving rise to random outputs $Y(u)$ and $Y(u')$.

- How can we tell which one is better?
- How can we find the best possible decision u ? What do we mean by the “best”?

- Deterministic vs statistical views -

Deterministic view vs uncertainty

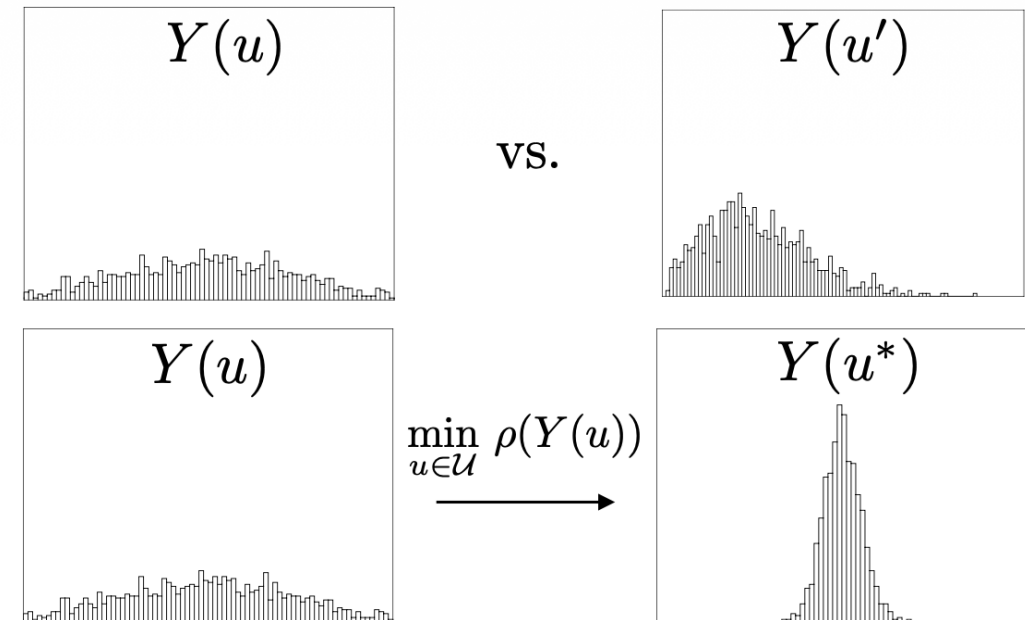
Deterministic view –

- $Y(u)$ and $Y(u')$ will have set values $y(u)$ and $y(u')$
- Compare values – choose the best option for u or u'

Statistical view –

- $Y(u)$ and $Y(u')$ will have multiple possible outcomes (RVs)
- **Concept of “better” – ambiguous.**
- Comparison is not possible.

How to compare RVs consistently – by using the PDFs/CDFs (as they fully characterize the RVs) or using risk measures (summarizing statistics)



Let's go back to our Gibbs reactor example -

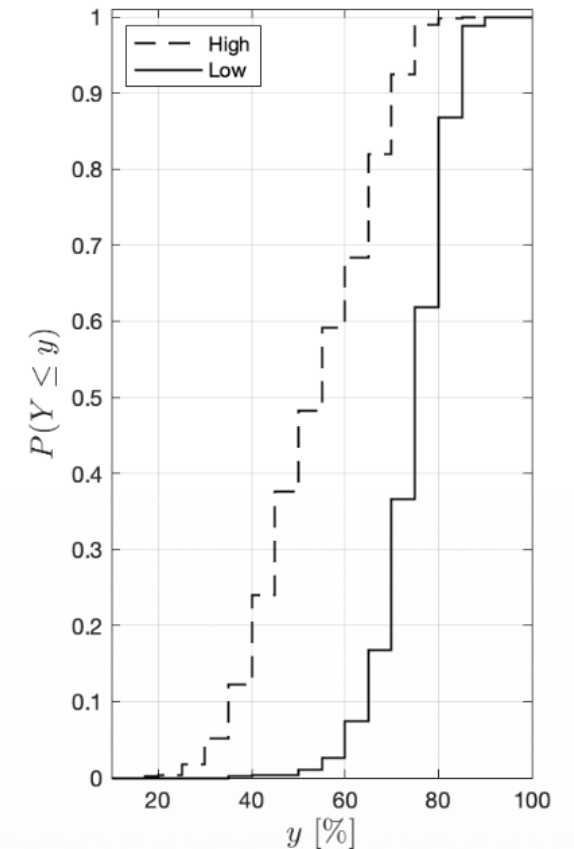
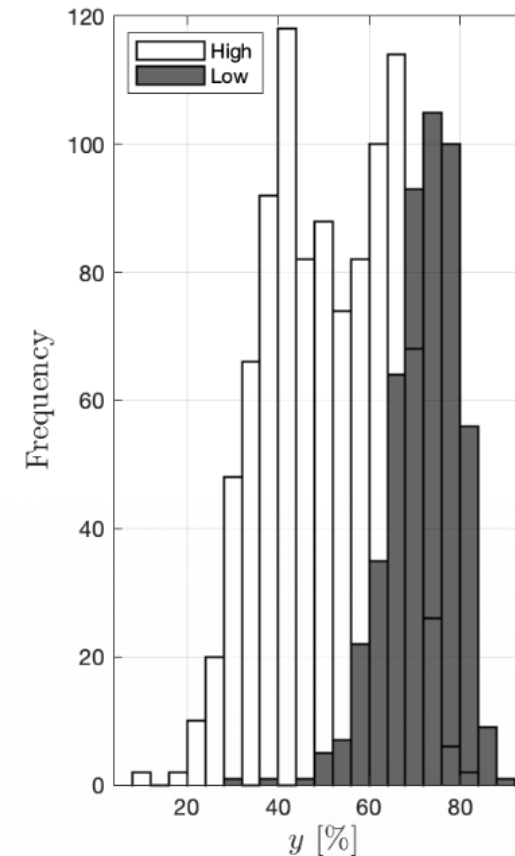
- Temperature – control variable. Can use it to counteract variability in pressure (X).

$$Y = \phi(X, u)$$

- **Question** – should we operate at lower or higher temperatures?
- **Goal** – maximize conversion or minimize variability?

Let's go back to our Gibbs reactor example

- $u \in [583, 613]$ – possible decisions to choose from.
- Lower temperature – higher conversion.
- If we want to maximize conversion, then we select lower temperature. Makes sense for exothermic reaction.
- Almost 0 probability that conversion will be lower than 40% at lower temperature.
- **What to select for reducing variability?**

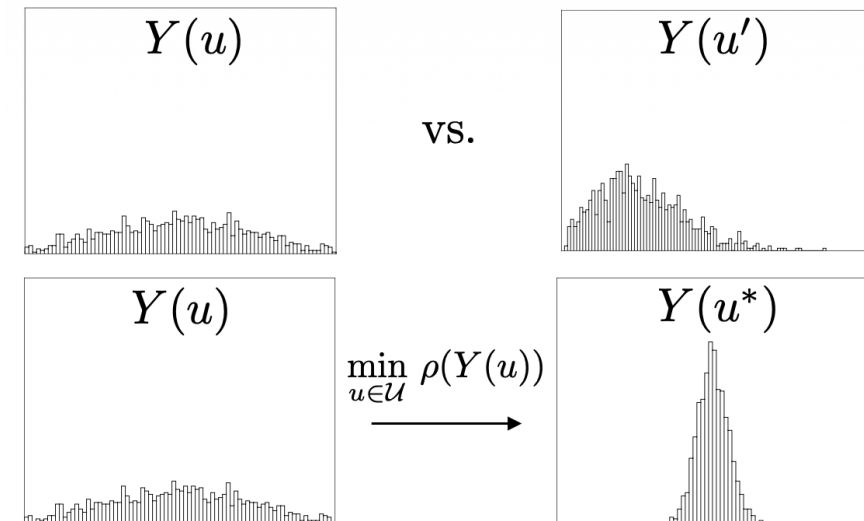


Decision making under uncertainty -

- Decisions (u) can shape the output of interest, making it less susceptible to uncertainty.
- Eg. Choose experimental conditions that reduce variance?
- Decision making – we might not be interested in comparing decisions but choosing the best possible decision.
- To do so, we select a measure of the output variable ($Y(u)$).

$$\min_{u \in \mathcal{U}} \rho(Y(u))$$

- Stochastic optimization



Conclusions

- Deterministic vs statistical view
- Random variables
- Summarizing statistics
- Uncertainty/variability with RVs.
- **Data-models-decisions** pipeline